

Field Relationships and Geochemical Constraints on the Emplacement of the Jinchuan Intrusion and its Ni-Cu-PGE Sulfide Deposit, Gansu, China

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Abstract

Field mapping and petrological-geochemical investigation of the Jinchuan intrusion in north-central China clarifies how the intrusion was emplaced and provides a new model that explains how its large and rich Ni-Cu-platinoid deposits may have formed. The intrusion was emplaced into high-grade gneisses and marbles along a disconformity at the base of an overlying cover sequence, indicating that it was emplaced as a sill, not a near-vertical dike, as previously proposed. After emplacement the intrusion was rotated to its present orientation and deformed and metamorphosed under greenschist-facies conditions. Relative enrichment of incompatible trace elements coupled with negative U-Th and Nb-Ta anomalies in all samples from the intrusion provide evidence that the parental magma assimilated granitoid rocks in the lower crust. The presence of abundant marble xenoliths, now decarbonized to diopside-rich skarns, and chemical indices such as high CaO/SiO₂, indicate that the magma assimilated carbonate on reaching its present site. This contamination may be linked to the formation of the Ni-Cu platinoid ores. We propose that the assimilation of carbonate-rich fluids increased the oxygen fugacity of the magma and led to the segregation of metal-rich sulfides.

Introduction

MOST MAGMATIC Cu-Ni sulfide deposits are hosted in lava flows or intrusions of mafic to ultramafic composition. The largest deposits, in Sudbury, Canada, are unique in that they are the only ones linked to a meteorite impact; others formed during the emplacement of mantle-derived magmas of diverse compositions in contrasting tectonic settings. The Kambalda deposits in Western Australia are located in komatiite lava flows in Archean greenstone belts, the Noril'sk-Talnakh deposits in Russia occur in near-surface intrusions related to the Siberian flood basalts, and the Voisey's Bay deposit in Labrador is in intrusions with an anorthosite magmatic affinity. A common element of the last three deposits is an association with S-rich country rocks and the assimilation of these rocks is assumed to have played an important role in the ore-forming process. The nature of the assimilant varied (i.e., pyrite-rich chert at Kambalda, anhydrite at Noril'sk, and S-rich metasediments at Voisey's Bay), but in each case assimilation of external sulfur is thought to have triggered the segregation of metal-rich sulfides (e.g., Naldrett, 1992; Li and Naldrett, 1999; Leshner and Keays, 2002).

The Jinchuan deposit in northern China is the third largest magmatic Ni-Fe sulfide deposit, with an estimated 500 million metric tons (Mt) of ore, grading 1.2 percent Ni and 0.7 percent Cu (Barnes and Tang, 1999). It differs from the other major deposits in several important ways. It is hosted

by amphibolite facies para- and orthogneisses and marbles and may have formed deeper in the crust than the deposits described above (Tang, 1993; Tang and Li, 1995). More significantly, the country rocks surrounding the intrusion are conspicuously poor in S-bearing minerals, and it seems unlikely that assimilation of external sulfur triggered ore formation.

In this paper, we present the results of a field and petrologic-geochemical study of the intrusion that hosts the Jinchuan deposit. We focused our attention on the rocks surrounding the intrusion in order to establish (1) the conditions of intrusion, including depth within the crust and character of wall rocks at the time of intrusion; (2) the form of intrusion, whether it was concordant or discordant; (3) the nature of the magma, its chemical composition and physical state; and (4) the manner in which the magma interacted with wall rocks. On the basis of our observations and results we propose a new model for the formation of the deposit that does not involve the assimilation of S-rich wall rocks.

Geologic Setting

The Jinchuan intrusion is located in north-central China (Fig. 1) at the northern margin of the elongate, fault-bounded Longshoushan thrust belt. The belt is mainly composed of Proterozoic high-grade ortho- and paragneisses and marbles of the Baijiazui and overlying Tamazigou Formations (Fig. 2), invaded by multiple generations of granitic and mafic dikes and veins (Figs. 1–3). A less metamorphosed cover sequence unconformably overlies the gneisses. This unit

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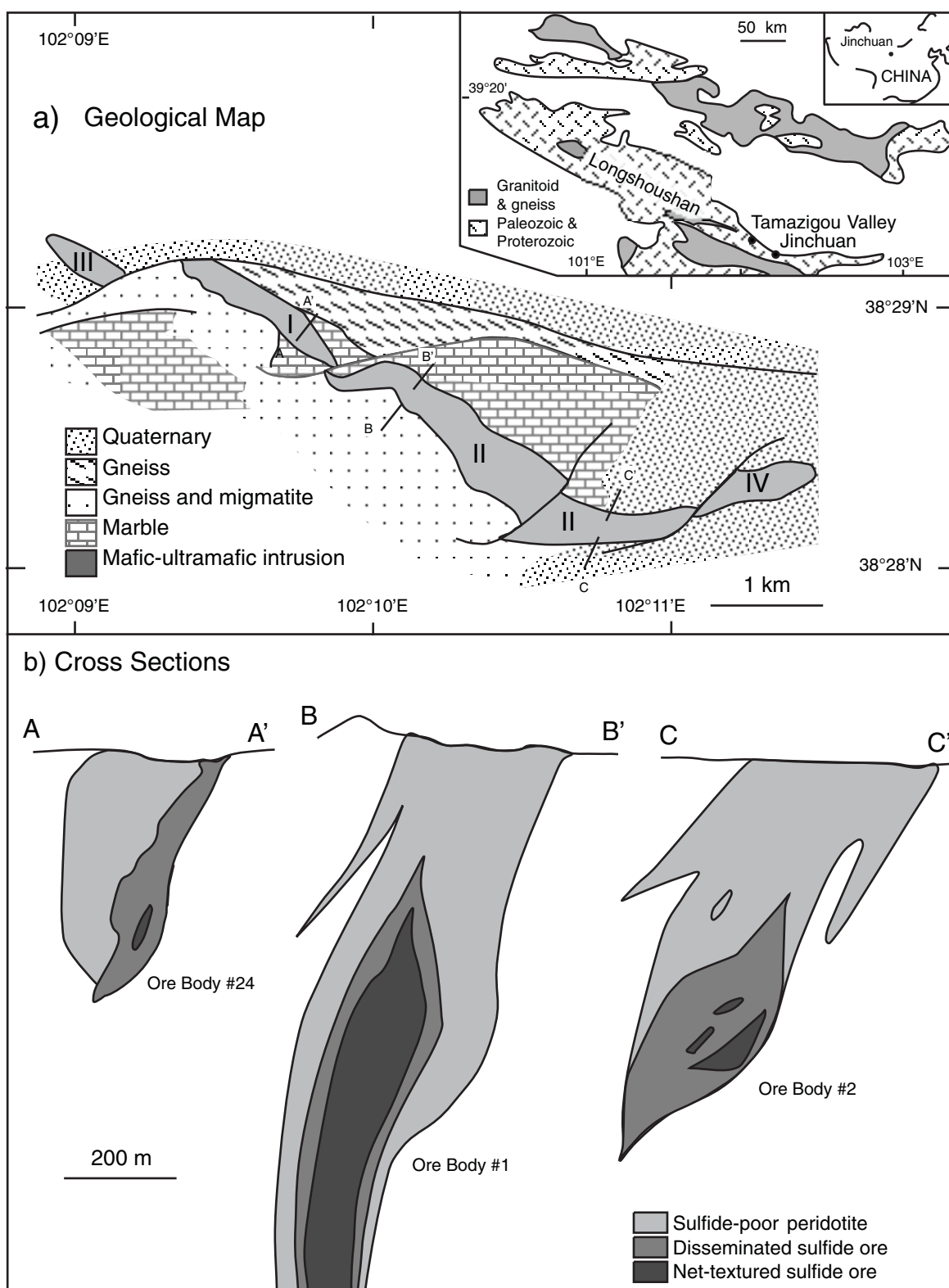


FIG. 1. (a). Geologic map of the Jinchuan intrusion and surrounding rocks. The Roman numerals identify four of five mining districts. The insets show the location within China and the geology of the Longshoushan belt. The position of the Tamazigou section is shown in the larger inset. (b). Cross sections through the Jinchuan intrusion. The map was originally from Tang (1993) but has been modified by Chai and Naldrett (1992a), with additional detail added here from an unpublished 1:1000 map of Jin Ao, Jinchuan Mining Company.

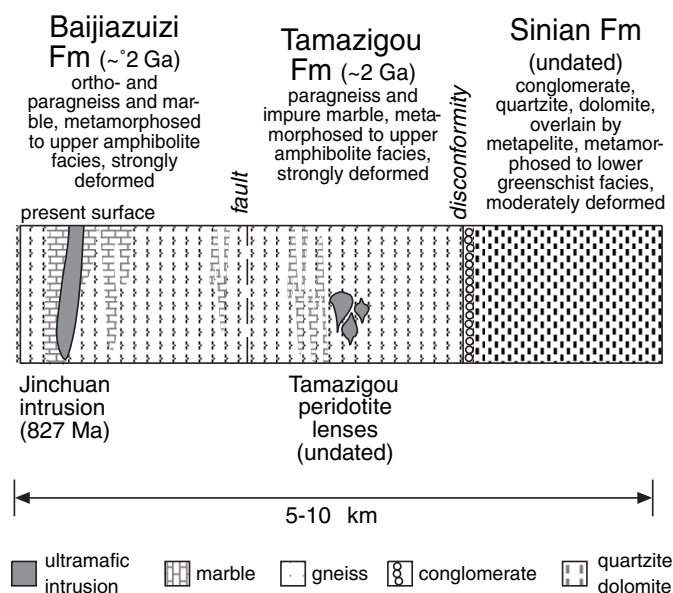


FIG. 2. Schematic section showing the stratigraphy of the northeastern part of the Longshouhan belt, as exposed along the Tamazigou Valley, and the approximate locations of the Jinchuan intrusion and Tamazigou peridotite lenses. The Jinchuan intrusion is oriented approximately parallel to the near-vertical disconformity at the base of the Sinian Formation. Rotation of the section brings both the disconformity and the intrusion to a horizontal orientation.

comprises the Denzigou conglomerates and quartzites, metabasalts, shales, and limestones and the younger Hanmushan sericite-quartz schists, calcareous schists, and limestone

breccias (Chai and Naldrett, 1992a, b; Tang, 1993; Tang and Li, 1995; Zhou et al., 2002; X. Li et al., 2004).

A U-Pb zircon age of ~2 Ga obtained by Tang and Li (1995) on a granitic dike that crosscuts Baijiazui gneisses provides a minimum age of the Paleoproterozoic formations, the dominant component of the Longshouhan belt. The metasedimentary cover sequences are believed to be of Sinian (Middle to Upper Proterozoic) age.

Some controversy surrounds the age of the Jinchuan intrusion. Until recently it was assigned a Middle Proterozoic age, based on a series of geochronological studies. K-Ar dating of biotites from the intrusion yielded ages of 1505 and 1529 Ma (Jia, 1986), Rb-Sr dating of biotite from a lamprophyre dike that cuts across the intrusion gave an age of 1339 Ma, and Tang et al. (1992) reported a whole-rock Sm-Nd isochron age of 1508 ± 31 Ma. U-Pb SHRIMP dating of zircon and baddelyite extracted from two plagioclase ilmenite from the intrusion and of zircon from a dolerite dike cutting across the intrusion gave more reliable results (X. Li et al., 2004, 2005). In both cases ages ranging from 880 Ma to 2.4 Ga were obtained from zircon xenocrysts, but the important constraint came from euhedral, zoned zircons, interpreted as magmatic and giving an age of 827 ± 8 Ma. This result is within the rather large error of the baddelyite age of 812 ± 25 Ma. Zircon from the crosscutting dolerite dike gave an age of 828 ± 3 Ma. Together, these results provide a convincing argument that the crystallization age of the intrusion was Upper Proterozoic, around 827 Ma.

The Jinchuan Intrusion and Ore Deposits

The intrusion, as currently exposed, consists of five fault-bounded lenses with a total length of 6,500 m and a maximum

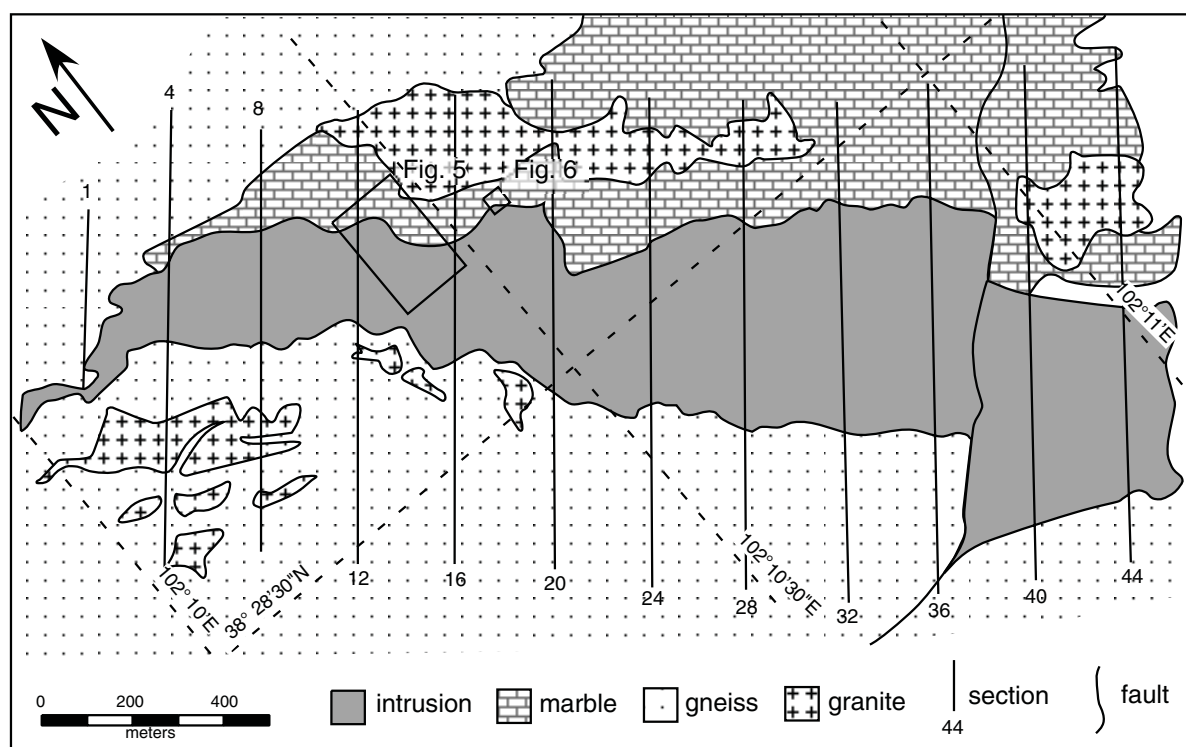


FIG. 3. Map of the main part of the Jinchuan intrusion in Mining District II (simplified from an unpublished 1:1000 map of Jin Ao, Jinchuan Mining Company), showing locations of detailed maps and sections (Figs. 5, 6, 11).

width of about 530 m (Figs. 1, 3; Chai and Naldrett, 1992a, b; Tang and Li, 1999). The northeast part of the intrusion can be traced through mining activities to a depth of at least 1,100 m. The intrusion strikes 140° and dips 50° to 80° SW. Although described as “trumpet shaped” in cross section in many publications (Chai and Naldrett, 1992a, b; Tang and Li, 1995), mining company maps and our own mapping demonstrate that the intrusion comprises a series of en echelon fault-bounded tectonic lenses truncated by the present erosion surface.

The outstanding petrologic characteristic of much of the intrusion is its extreme homogeneity. The interior of the central part, referred to as Mining District II in the terminology used by the mine geologists (Figs. 1, 3), is composed entirely of coarse-grained olivine cumulates that are variably altered and deformed but whose magmatic character varies little. The intrusion is well exposed, both in outcrop and in mine workings, yet despite careful inspection we found no trace of the shallow-dipping magmatic layering described by Chai and Naldrett (1992a). Facies rich in ortho- and clinopyroxene and plagioclase are developed at both margins and toward the northern end of the intrusion, but changes from olivine-rich to olivine-poor rocks are gradual and no obvious magmatic contacts were observed.

Magmatic minerals are variably well preserved throughout the intrusion (Fig. 4); in some places alteration is negligible,

in others all primary phases have been transformed into an assemblage of serpentine, tremolite-actinolite, and chlorite (Tang and Li, 1995).

The ore deposits are mainly restricted to the north-central part of the intrusion where they occur as elongate lenses oriented parallel to its steep margins, particularly on the marble side (Fig. 1). They consist of abundant (1–10 vol %) Cu-Ni-Fe sulfides that usually form a continuous matrix between large olivine grains (net texture). Chai and Naldrett (1992a, b), C. Li et al. (2004), and de Waal et al. (2004) provided more detailed descriptions of the deposits.

Methods

Our fieldwork focused on the petrology and contact relationships between the Jinchuan intrusion and its host rocks, at both local and regional scales. We mapped several areas along the northern contact of the intrusion at scales between 1:300 and 1:1000 (Figs. 5–6), and we also traversed and sampled a section along the Tamazigou Valley which cuts across the Longshoushan belt about 10 km northwest of the Jinchuan mine. We collected 45 samples and analyzed them petrographically by scanning electron microscope and electron microprobe, and for major and trace elements and carbon and oxygen isotopes.

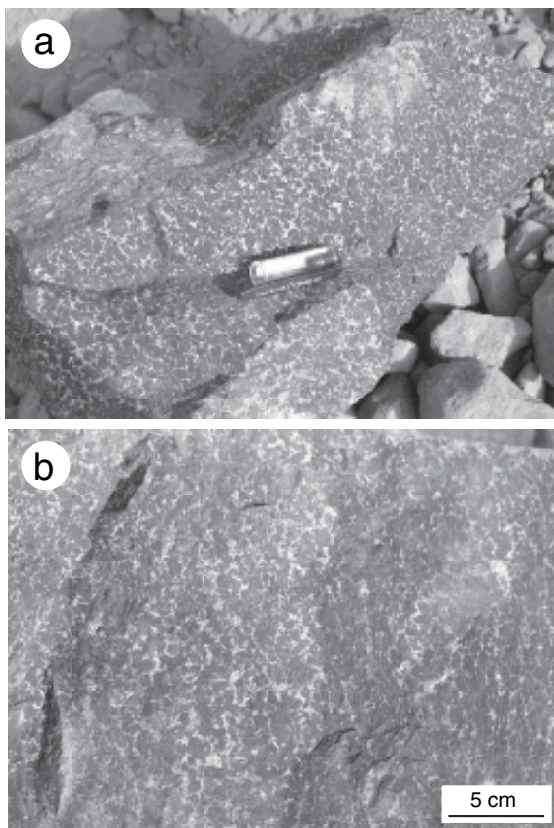


FIG. 4. (a). Peridotite with well-developed magmatic texture. Cumulus, dark-colored grains of serpentinized grains and pyroxene lie in a plagioclase-phlogite-rich matrix. (b). Net-textured ore in which dark-colored grains of serpentinized grains of olivine are enclosed in a matrix of Ni-Cu-Fe sulfides. The form and the size of the olivine grains are similar in both samples; only the matrix is different.

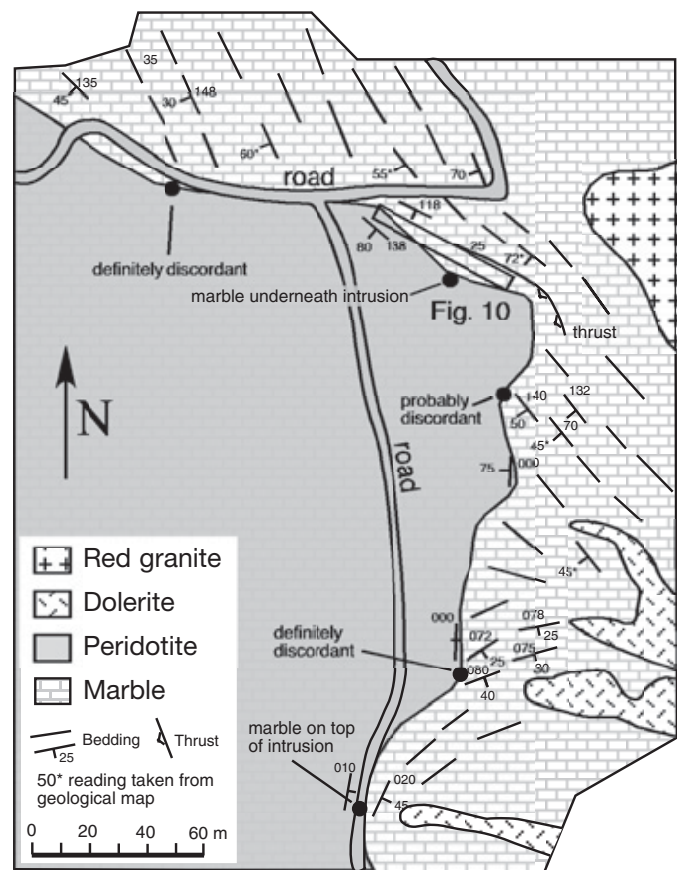


FIG. 5. Detailed sketch map of part of the northern contact of the Jinchuan intrusion, showing the locally discordant but regionally concordant relationship between the intrusion and its host rocks. The location of this map is shown in Figure 3. The inclined rectangle at the contact shows the location of the section reproduced as Figure 10.

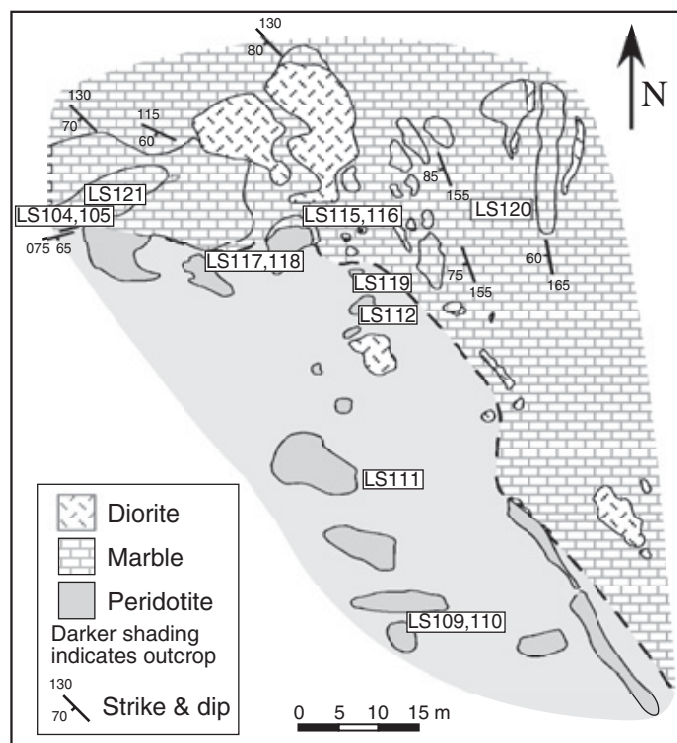


FIG. 6. Detailed sketch map of another part of the northern contact of the Jinchuan intrusion. This map shows the locally discordant nature of the contact and the positions of samples of intrusive rocks and marbles. The location is shown in Figure 3.

The major and transition elements were measured at the Centre de Recherche Petrologique et Geochemique in Nancy, France (see www.crpq.cnrs-nancy.fr/SARM/ for analytical procedures and estimates of precision); trace elements were measured by inductively coupled plasma mass spectrometry at the Laboratoire de Géodynamique des Chaînes Alpines in Grenoble, using the method of Ionov (2004). This method produces accurate data for trace elements present at low concentrations in ultramafic rocks; the uncertainties are estimated at better than 5 percent for most elements reported in Table 1, except for those at the parts per million level for which the uncertainty is around 10 percent. A scanning electron microscope at the University of Chambéry was used for preliminary chemical mapping of mineral phases and limited quantitative analysis was undertaken with an electron microprobe at Université Claude Bernard in Lyon. These results largely complement the more abundant data reported by Chai and Naldrett (1992a).

All stable isotope measurements were made at the University of Cape Town. Oxygen isotope ratios of whole rocks were determined using conventional extraction methods after reaction with ClF_3 . The O_2 was converted to CO_2 using a hot platinized carbon rod, and isotope ratios were measured using a Finnigan MAT Delta XP mass spectrometer in dual-inlet mode and are reported in the familiar δ notation, where $\delta = (R_{\text{sample}}/R_{\text{standard}} - 1) \times 1,000$ and $R = {}^{18}\text{O}/{}^{16}\text{O}$. The quartz standard NBS28 was run twice with each batch of eight samples and the raw data were normalized to a value of NBS28 of 9.64 per mil. The difference between normalized

and unnormalized data is <0.5 per mil. Carbon and oxygen isotope ratios on carbonate-bearing samples were determined after reaction of 20 mg of powdered sample and 5 ml of 100 percent H_3PO_4 at 25°C . Samples were tested for further reaction at 50°C , but yields were negligible suggesting that little or no dolomite is present. The CO_2 extracted from the calcite samples was analyzed for both carbon and oxygen and the data were corrected using the CO_2 -calcite fractionation factor of 1.01025. An in-house carbonate standard, Namaqualand marble (NM), was run with every five samples. The data obtained were used to convert the raw data to the PDB and SMOW scales. The estimated precision for carbonate analyses based on repeated measurement of duplicate standards was better than ± 0.1 and ± 0.2 per mil for $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$, respectively, and for silicates the estimated precision, based on repeated duplicate analyses of NBS-28, was ± 0.2 per mil (Table 2).

Structural and Metamorphic Relationships

Several observations emerged from our study of the metamorphic rocks of the Longshoushan belt.

The metamorphic sequence to the northeast of the intrusion consists of olivine marbles with distinctive bedding and metamorphic mineralogy (Figs. 3, 5–6). Bedding is well preserved, particularly farther from the intrusion, as layers of variable thickness (20 cm to 1 m), contrasting grain size and, locally, different mineralogy. In places near the contact of the intrusion, the marbles are cut by several generations of sheared and unsheared amphibolite (originally dolerite) dikes (Figs. 5–6).

The sequence to the southeast of the Jinchuan intrusion mainly comprises migmatitic, banded, hornblende-biotite gneisses in which isoclinally and pygmatically folded granite veins cut the regional biotite-sillimanite foliation and paleosome layers and lenses. Such anatexitic structures and metamorphic mineral assemblages indicate recrystallization under upper amphibolite facies conditions above the sillimanite and partial melt isograds.

In the Tamazigou Valley, about 10 km northwest of the Jinchuan intrusion, gneisses of the Baijiazui and Tamazigou Formations contain layers and lenses of sillimanite-biotite schists, amphibolites several hundred meters thick, diopside marbles, biotite schists, and quartzites (Fig. 2). A distinctive, vertical 2-m-thick conglomerate marks the base of the overlying, vertical (unnamed) Sinian formation. This unit, which contains well-rounded pebbles of quartzite and quartz in a quartz matrix (Fig. 7), is conformable with biotite schists below and stratigraphically overlain conformably by a succession (more than 1 km thick) of well-bedded quartzites, interbedded quartzites and marbles, muscovite-quartz schists, and chlorite-biotite schists (Figs. 2, 7). There is no discernible deformation at the base of the conglomerate. The quartzites contain crossbeds that are the right way up with respect to the position of the conglomerate underneath. The upper rocks are weakly deformed with a locally developed chloritic cleavage that is commonly bedding-parallel and strikes 130° to 140° , subparallel to the foliation in gneisses of the Tamazigou Formation immediately underneath. The metamorphic grade of these rocks, estimated from the mineralogy and textures in the metapelites, is greenschist facies. We interpret these

TABLE 1. Locations and Brief Petrographic Descriptions of Samples

Sample no.	Lithology	Primary mineralogy	Locality	
LS 101	Xenolith-peridotite contact	Olivine orthocumulate/granular (1-2 mm) diopside	On road near mine 3	N38°27'90" E102°10'36"
LS 102	Diopside xenolith	Pale dirty green colored granular diopside	On road near mine 3	N38°27'90" E102°10'36"
LS 103	Diopside xenolith	Pale dirty green colored granular diopside	On road near mine 3	N38°27'90" E102°10'36"
LS 108	Ol cumulate, cpx-plag-biot matrix	50% large (3 mm) olivine, 40% px, 5-10% plag-biot matrix	Detailed map (Fig. 6)	~N38°28' E102°10'37" Grid reference 60S 20E
LS 109	Ol cumulate, cpx-plag-biot matrix	45% olivine, 45% px, 5-10% plag-biot matrix	Detailed map	41S 10E - resistant band
LS 110	Ol cumulate, cpx-plag-biot matrix	50% olivine, 40% px, 10% plag-biot matrix	Detailed map	40S 10E -resistant band
LS 111	Ol cumulate, cpx-plag-biot matrix	50% olivine, 40% px, 5-10% plag matrix, some sulfide	Detailed map	25S 0E
LS 112	Ol cumulate, cpx-rich matrix	70% olivine, 30% matrix	Detailed map	5S 5E
LS 113	Ol cumulate, plag-rich matrix	50% olivine, 40% px, 5-10% plag-biot matrix	Detailed map	0S 0E
LS 114	Ol cumulate, plag-rich matrix	50% olivine, 40%px, 10% calcite-rich matrix	Detailed map	0S 0E
LS 115	Olivine marble	30% small (~1 mm) euhedral serpentinized olivine, coarser granular calcite	Detailed map	2S 2E
LS 116	Olivine marble	30% small serpentinized olivine grains, coarse granular calcite	Detailed map	2S 2E
LS 117	Olivine marble	40% olivine, coarse granular calcite	Detailed map	2S 2E
LS 118	Olivine marble	5% olivine-tremolite-Fe oxide in marble	Detailed map	2S 2E
LS 119	Ol cumulate with carbonate matrix	Olivine-pyroxene marble	Detailed map	3S 4E
LS 120	Mica-rich marble	Marble with 10% white mica	Detailed map	2N 24 E
LS 121	Olivine marble	30% olivine, coarse granular calcite	Detailed map	7N 7W
LS 130	Olivine-opx orthocumulate	40% olivine, 20%px, 40% plag matrix	Section across intrusion	N38°28'61" E102°10'36"
LS 131	Olivine orthocumulate	60% olivine (altered to brown serpentine), 10% px, 30% matrix	Section across intrusion	N38°28'59" E102°10'33"
LS 132	Olivine orthocumulate	60% olivine (altered to brown serpentine), 10% px, 30% matrix	Section across intrusion	N38°28'59" E102°10'31"
LS 133	Olivine-opx orthocumulate	Olivine cumulate-completely serpentinized	Section across intrusion	N38°28'57" E102°10'33"
LS 134	Olivine adcumulate	Olivine cumulate-brown and blue-green serpentine	Section across intrusion	N38°28'55" E102°10'33"
LS 135	Olivine-opx adcumulate	Cumulus olivine, large cpx oikocrysts-strongly serpentinized	Section across intrusion	N38°28'58" E102°10'19"
LS 136	Opx cumulate	Cumulus olivine (serpentinized), large cpx unaltered oikocrysts	Section across intrusion	N38°28'65" E102°10'11"
LS 137	Olivine-opx orthocumulate	60% olivine, 15% cpx oikocrysts	Section across intrusion	N38°28'61" E102°10'11"

Abbreviations: biot = biotite, ol = olivine, opx = orthopyroxene, plag = plagioclase, px = pyroxene

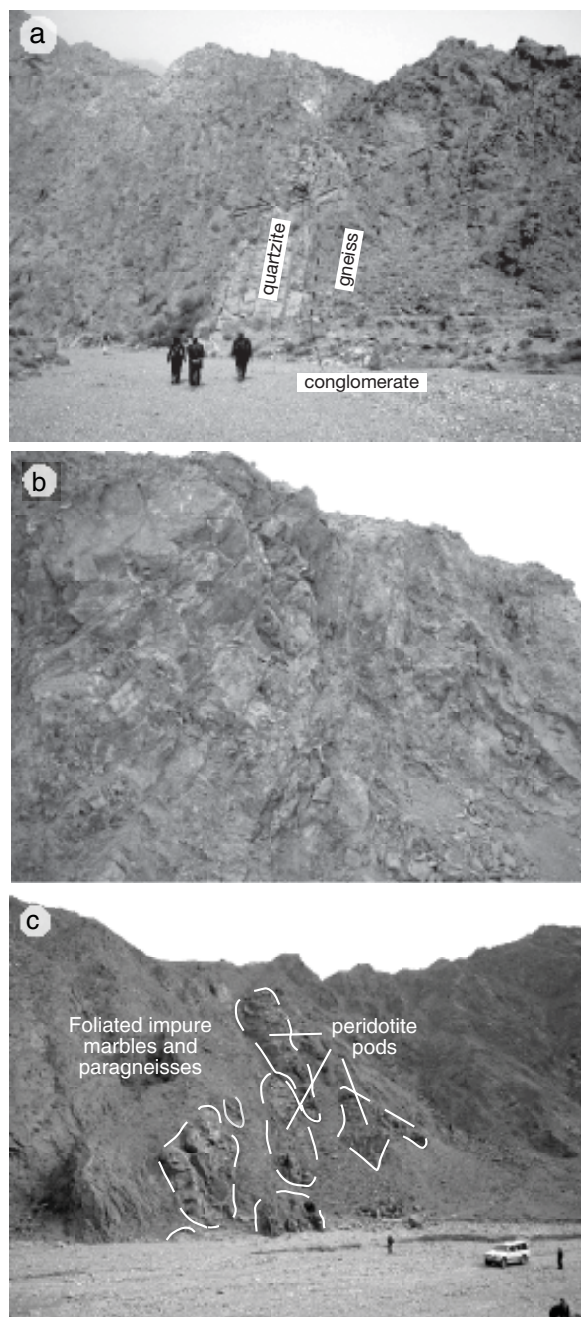


FIG. 7. (a). Steeply dipping unconformity between Proterozoic quartzite (to left = south) and felsic gneisses in the Tamagouzi valley about 20 km northwest of Jinchuan ($38^{\circ}30'40''$ N, $102^{\circ}02'51''$ E). (b). Thrust zone in felsic gneisses at the southeast end of the Jinchuan intrusion, Mining District II. (c). Photo of peridotite pods (boudins) in highly sheared felsic gneiss and marble from the Tamagouzi valley (N $38^{\circ}30'04''$, E $102^{\circ}04'56''$).

relationships in terms of a high-grade basement overlain conformably by a basal conglomerate and low-grade cover. Rocks above the conglomerate are well-bedded sediments, but those below it are strongly foliated schists and gneisses. The metapelites demonstrate the different metamorphic grades: sillimanite-biotite below versus chlorite-muscovite-biotite above. Thus the conglomerate marks a major metamorphic and structural boundary at a disconformity that separates older rocks

that were essentially horizontal (after presumed erosion) at the time of deposition of the upper rocks.

Conspicuously absent in the country rocks surrounding the Jinchuan intrusion are sulfur-rich rocks such as sulfide-bearing sediments or anhydrites, the types of rock that are thought to provide external S sources during the formation of Ni-Cu sulfide deposits at Noril'sk-Talnakh (Lightfoot and Hawkesworth, 1997; Naldrett, 2004) and Kambalda (Leshner and Keays, 2002).

As Figure 7a shows, the strike of the intrusion and host gneisses is close to that of the disconformity, but, on a larger scale, the intrusion and gneisses have a moderate to steep dip to the south, whereas the cover sediments are subvertical. If the Jinchuan intrusion had invaded the gneisses at or before the time of deposition of the cover sediments, as suggested here, its initial orientation would have been close to horizontal. The implication, discussed in more detail later in the paper, is that the Jinchuan intrusion was originally a shallow-dipping sheet and not a vertical funnel- or dike-like intrusion, as described or inferred by Tang and Li (1995, 1999) and Chai and Naldrett (1992a, b).

Midway along the Tamazigou Valley, ~10 km northwest along strike from Jinchuan at $38^{\circ}30'12''$ N, $102^{\circ}04'60''$ E, deformed paragneisses, schists, and marbles of the Tamazigou Formation contain small boudinaged lenses of peridotite up to 30 m across (Figs. 2, 7c, d). The lenses have cores of olivine-pyroxene cumulate (now altered to serpentine and amphibole) and margins of actinolite schist and chlorite schist. Intercalated layers of pure or diopside-bearing marble, quartzite, sillimanite-garnet mica schist, and diopside gneiss surround the lenses. The foliation in the margins is conformable with that in the surrounding rocks.

Contact Relationships of the Intrusion

Northern footwall contact

The country rocks immediately to the north of the intrusion consist mainly of marble. The dominant mineral assemblage is calcite and olivine (now altered to serpentine) with minor diopside clinozoisite, tremolite, and hydrogrossular, the product of metamorphism of dolomitic limestone. Many dikes of amphibolite and a few of pegmatite and red granite intrude the marble, and drill cores that cross the contact with the intrusion contain gneiss intercalated with marble. In the marble, the bedding is well preserved, with layers of variable thickness (20 cm to 1 m), contrasting grain size and, locally, different mineralogy. The rocks strike 130° to 140° and dip steeply to the south. The bedding of the marbles is generally concordant with the northern margin of the Longshoushan belt and, on a smaller scale, with the northern contact of the intrusion. However, in low strain zones at major bends the contact of the intrusion is markedly discordant to the bedding of the marbles, as shown in detail in Figure 5.

Although the metamorphic grade (olivine marble) is similar at the contact with the intrusion and at a distance of 300 m to the north of it, there are three indications of the presence of a contact metamorphic aureole.

1. The abundance of olivine (now altered to serpentine) within the bulk rock increases toward the intrusion, inversely

TABLE 2. Major and Trace Elements and Oxygen Isotope Compositions of

Locality Sample no.	Detailed map (Fig. 5) LS119	Detailed map LS120	Detailed map LS115	Detailed map LS116	Detailed map LS117	Detailed map LS118	Detailed map LS121	Northern margin LS103	Northern margin LS102A	Northern margin LS102B	Section LS131	Section LS132
Percent	Marble	Mica marble	Olivine marble	Olivine marble	Olivine marble	Olivine marble	Olivine marble	Xenolith	Xenolith	Xenolith	Dunite	Dunite
SiO ₂	31.68	26.21	15.2	16.08	18.33	4.34	12.95	49.23	53.75	54.72	36.88	35.39
TiO ₂	0.81	0.14	<dl	0.03	0.05	<dl	0.06	0.17	<dl	<dl	0.28	0.18
Al ₂ O ₃	10.07	3.19	0.82	0.27	2.29	1.07	0.98	2.08	0.31	0.97	3.08	2.4
Fe ₂ O ₃ (total)	6.74	0.93	3.25	1.72	3.58	1.8	2.89	5.72	2.22	4.53	13.48	13.44
MnO	0.17	0.03	0.08	0.1	0.1	0.1	0.1	0.11	0.1	0.1	0.16	0.12
MgO	18.79	20.71	15.38	16.42	19	4.89	16.06	21.2	19.37	22.68	32.61	33.4
CaO	19.68	23.35	33.72	33.68	28.2	47.59	31.63	16.85	21.06	13.15	1.48	0.34
Na ₂ O	<dl	<dl	<dl	<dl	<dl	<dl	<dl	0.27	<dl	0.31	<dl	<dl
K ₂ O	<dl	1.54	<dl	<dl	<dl	<dl	<dl	<dl	<dl	<dl	<dl	<dl
P ₂ O ₅	0.05	0.05	<dl	<dl	0.04	<dl	<dl	0.03	<dl	0.03	0.09	0.1
LOI	11.97	22.87	31.46	31.91	28.14	39.05	33.45	4.16	1.62	2.35	11.27	12.93
Total	99.97	99.01	99.92	100.22	99.72	98.83	98.12	99.82	98.43	98.84	99.33	98.3
Ppm												
Cr	50	25.0	<dl	<dl	<dl	22	<dl	1244	<dl	77.7	5610	7941
Ni	33.0	2.6	65.7	71.7	194	90.3	18.6	805	113	15	1351	1534
Cu	85.2	<dl	62.8	30.7	44.7	110	<dl	481	222	311	13.6	16.4
V	84.1	56.5	21.9	16.1	27.9	26.8	37.1	54.4	17.6	25.9	59.1	45.7
Cs	0.06	3.32	0.04	0.04	0.03	0.01	0.03	0.3	0.17	0.06	1.76	1.37
Rb	0.31	48.2	0.23	0.37	0.22	0.01	0.16	1.16	1.33	0.63	3.73	1.91
Ba	19.5	3733	23.9	60.2	32.4	16.6	24.7	54.0	12.4	52.3	74.2	17.6
Th	2.55	0.83	0.58	0.97	0.83	0.25	0.33	0.42	0.32	0.44	0.4	0.23
U	1.26	4.44	4.66	1.09	5.6	1.26	3.53	0.25	0.36	0.7	0.07	0.07
Nb	5.82	3.01	0.34	1.22	1.31	0.1	0.98	0.94	0.32	0.58	1.48	0.82
Ta	0.85	0.21	0.03	0.08	0.08	0.01	0.14	0.06	0.05	0.09	0.08	0.04
La	9.69	6.95	18.4	13.0	29.6	2.62	1.79	1.31	0.81	0.54	1.77	1.14
Ce	25.1	13.11	36.5	26.5	49.4	7.38	4.39	4.26	2.11	2.29	5.09	3.56
Pr	3.49	1.57	3.87	3.14	4.93	0.78	0.57	0.68	0.3	0.42	0.76	0.56
Pb	2.62	4.05	0.77	1.44	42.7	4.82	1.98	51.2	5.83	1.84	1.89	1.58
Nd	14.5	5.55	14.1	10.8	15.7	3.4	2.45	3.1	1.09	1.81	3.52	2.49
Sr	40.1	231	150	195	147	144	83.2	79.8	30.4	12.3	38.3	12.3
Sm	3.19	0.92	2.35	1.75	2.3	0.7	0.53	0.77	0.24	0.41	0.84	0.59
Zr	100	29.6	0.49	2.42	8.53	0.74	19.3	18.2	3.57	7.0	26.9	15
Hf	0.8	0.77	0.02	0.04	0.28	0.03	0.53	0.49	0.11	0.24	0.61	0.37
Eu	0.76	0.32	0.46	0.23	0.47	0.09	0.07	0.26	0.07	0.14	0.23	0.18
Gd	3.14	0.76	2.06	1.46	1.95	0.81	0.5	0.71	0.19	0.4	0.79	0.55
Tb	0.53	0.12	0.3	0.21	0.28	0.13	0.09	0.12	0.03	0.06	0.13	0.09
Dy	3.25	0.73	1.73	1.21	1.6	0.8	0.5	0.68	0.19	0.36	0.77	0.54
Ho	0.67	0.14	0.38	0.26	0.35	0.19	0.11	0.14	0.04	0.08	0.16	0.11
Er	1.95	0.44	1.02	0.78	0.97	0.57	0.33	0.39	0.13	0.2	0.45	0.33
Y	18.9	4.58	12.6	8.98	11.1	6.48	3.3	4.04	1.45	2.32	5.04	3.4
Yb	1.93	0.41	1.03	0.65	0.88	0.57	0.29	0.33	0.13	0.2	0.43	0.27
Lu	0.29	0.07	0.16	0.09	0.12	0.09	0.04	0.04	0.02	0.03	0.07	0.05
Per mil												
$\delta^{18}\text{O}$ silicate	na	na	na	na	na	na	na	na	13.5	9.53		
$\delta^{13}\text{C}$ calcite	-1.2	-0.5	-1.1	-1	-0.96	-0.96	-0.8	-2.75	No CO ₂	No CO ₂		
$\delta^{18}\text{O}$ calcite	16.2	17.4	16.8	16.9	16.84	19.22	18.1	15.75				
δ calcite	5	9.4	100	71	50	87	16	4	0	0		

Notes: <dl = below detection limit, na = not analyzed

to the proportion of bands or patches of pure, olivine-free marble. Near the contact, pure marble beds are up to 10 to 20 cm wide, at 50 m from the contact they are 20 to 30 cm wide, at 150 m they are up to 1 m wide, and at 300 m they are 2 to 4 m wide.

2. The size of the olivine grains is greatest near the intrusion and finer farthest from it.

3. Bedding is less well defined near the contact than it is 300 m distant where there is a well-preserved and pronounced stratigraphy expressed by regular and continuous bedding units, variations in grain size, and the presence or absence of fine-grained olivine in alternating beds. Throughout the marbles some beds are enriched in muscovite, others in tremolite; these probably reflect changes in bulk-rock compositions.

Ultramafic Rocks from the Jinchuan Intrusion and Surrounding Marbles

Section LS133	Section LS134	Section LS130	Section LS135	Section LS136	Section LS137	Detailed map, Fig. 5 LS108	Detailed map LS109	Detailed map LS110	Detailed map LS111	Detailed map LS113	Detailed map LS114	Detailed map LS112
Dunite	Dunite	Plag- wehrlite	Wehrlite	Wehrlite	Wehrlite	Plag- biotite- wehrlite	Plag- biotite- wehrlite	Plag- biotite- wehrlite	Plag- biotite- wehrlite	Plag- dunite	Plag- dunite	Plag- wehrlite
35.1	34.5	43.69	38.25	38.04	36.55	42.41	42.79	42.72	42.62	26.32	25.9	34.27
0.15	0.16	0.4	0.28	0.24	0.24	0.63	0.57	0.48	0.47	0.04	0.23	0.8
1.98	1.66	6.62	3.52	2.8	2.8	6.15	6.63	7.1	7.68	5.34	7.05	9.02
13.57	13.38	12.22	13.15	14.01	13.34	12.95	13.02	12.36	12.05	8.14	6.55	8.8
0.15	0.17	0.17	0.16	0.15	0.16	0.17	0.19	0.17	0.16	0.2	0.29	0.18
34.41	34.27	26.44	33.12	34.52	33.79	26.43	25.72	25.18	24.66	26.75	27.85	20.74
0.29	0.85	5.38	2.25	1.71	1.29	4.46	5.2	5.29	5.3	13.07	12.57	17.65
<dl	<dl	0.95	0.35	<dl	<dl	0.83	0.82	0.9	0.98	<dl	<dl	0.19
<dl	<dl	<dl	<dl	<dl	<dl	0.52	0.49	0.45	0.56	<dl	0.11	<dl
0.1	0.09	0.1	0.09	0.11	0.11	0.1	0.09	0.08	0.08	0.05	0.04	0.05
12.82	13.36	2.22	7.3	7.95	10.32	5.7	5.37	5.76	5.73	19.46	19.72	8.38
98.57	98.44	98.58	98.62	99.53	98.69	100.34	100.89	100.48	100.28	99.37	100.3	100.08
8431	6618	3331	5496.97	4411.44	5951.91	3349.68	3004.7	3093.43	2993.16	<dl	<dl	<dl
1469	1505	955	1248	1223	1573	57	63	51	41	1464	254	263
11.7	14.1	43.6	30.7	16.4	158	1036	932	929	888	813	75.7	191
23.3	40.0	109	41.6	63.3	37.3	137	133	114	112	26.3	50.9	69.8
0.36	0.9	2.19	0.35	0.32	0.15	0.62	2.1	na	2.46	na	0.44	3.7
0.58	1.29	12.4	4.19	2.24	3.21	4.68	13.8	na	12.81	na	5.57	34.0
8.87	10.7	390	122	6.23	77.2	480	367	na	161	na	293	520
0.17	0.17	0.64	0.3	0.43	0.23	1.72	0.96	na	0.8	na	0.82	1.57
0.1	0.03	0.1	0.05	0.09	0.06	1.71	0.14	na	0.16	na	1.4	0.28
0.66	0.61	1.86	1.17	1.13	0.99	2.5	3.56	na	2.44	na	3.4	6.29
0.03	0.04	0.1	0.06	0.06	0.06	0.42	0.18	na	0.14	na	0.31	0.33
1.52	1.28	10.1	3.22	1.51	2.42	6.36	8.79	na	6.79	na	9.81	15.1
3.47	3.09	19	7.36	4.21	5.53	16.5	19.5	na	14.9	na	17.5	35.1
0.44	0.42	2.03	0.97	0.64	0.74	2.44	2.57	na	1.97	na	1.86	4.56
2.95	1.08	8.06	5.02	0.55	0.89	29.5	24.0	na	2.49	na	1.48	11.3
1.91	1.92	7.63	4.11	2.96	3.14	10.5	10.4	na	7.99	na	6.55	18.8
12.1	33.9	142	79.5	13.2	57.9	87.4	199	na	186	na	66.0	306
0.43	0.47	1.6	0.9	0.71	0.7	2.67	2.27	na	1.65	na	1.22	4.04
12.1	11.3	35.1	20.7	21.2	20.6	134	73.0	na	66.3	na	29.2	124
0.28	0.29	0.82	0.51	0.48	0.46	4.21	1.71	na	1.53	na	0.47	2.92
0.12	0.14	0.56	0.3	0.22	0.23	0.59	0.72	na	0.58	na	0.24	1.23
0.4	0.45	1.55	0.88	0.66	0.71	2.79	2.3	na	1.72	na	1	3.95
0.07	0.07	0.25	0.14	0.11	0.11	0.45	0.36	na	0.27	na	0.16	0.64
0.42	0.43	1.44	0.83	0.67	0.66	2.58	2.16	na	1.64	na	0.89	3.69
0.09	0.09	0.31	0.17	0.13	0.14	0.54	0.44	na	0.34	na	0.19	0.78
0.24	0.26	0.83	0.48	0.37	0.38	1.46	1.23	na	0.93	na	0.56	2.07
2.58	2.45	9.04	4.93	4.25	4.25	15.4	13.6	na	9.74	na	5.85	23.4
0.23	0.24	0.73	0.44	0.35	0.35	1.4	1.12	na	0.85	na	0.54	1.89
0.04	0.04	0.12	0.07	0.06	0.06	0.21	0.16	na	0.13	na	0.1	0.28
						5.86	5.41	4.66	8.85	9.97	9.46	10.86

Marble xenoliths are common along the northern margin and occur sporadically throughout the intrusion. They range up to 60 cm wide and at least 4 m long and have sharp contacts with the enclosing ultramafic rock. Their shapes are irregular, ranging from angular to rounded, and they appear to have been incorporated into the magma as solid fragments (Fig. 7). The samples we studied consist entirely of a

recrystallized mosaic of nearly pure Al-poor diopside with minor spinel (Fig. 8, Table 1); the original marble has been totally decarbonatized. In some places the ultramafic host adjacent to the xenolith is only slightly altered; elsewhere it is largely to completely replaced by actinolite and chlorite.

Interaction at another scale is revealed by our study of samples collected in the detailed map area (Fig. 6). Here the

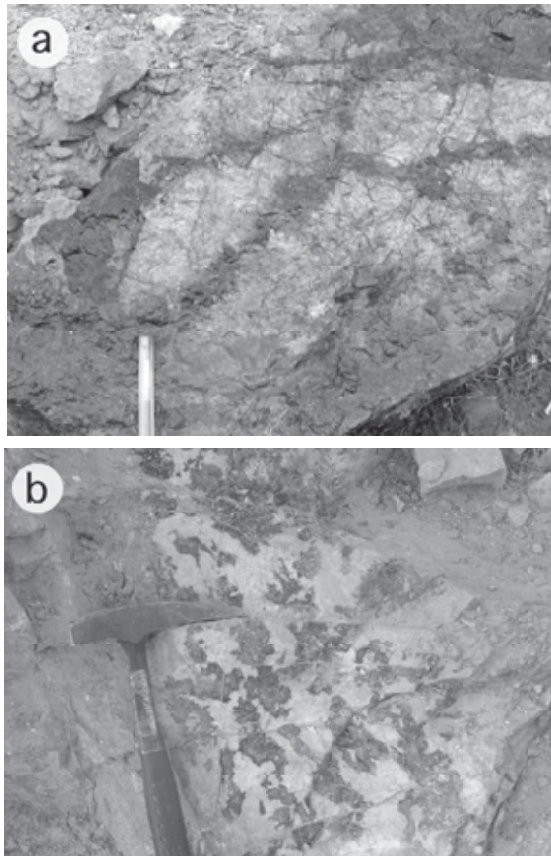


FIG. 8. (a). Xenoliths of marble, now altered to diopside skarn in peridotite near the northern margin of the intrusion. (b). Irregular patch of dark-colored peridotite mingled with light-colored marble at the northern margin of the intrusion ($38^{\circ}28'44''$ N, $102^{\circ}10'15''$ E).

contact is marked by a 2- to 3-m-wide zone of hybrid rocks; peridotites with carbonate-rich matrices on one side and marbles containing patches of peridotite on the other. Some hybrid rocks from the marble side of the contact zone contain grains of olivine and pyroxene with the form and size of the magmatic grains of the intrusion, either isolated or in clumps, within a carbonate-rich matrix. In one outcrop of marble about 2 m from the contact (sample LS119), 15 to 30 vol percent of patches and isolated grains of olivine are distributed irregularly within a more abundant calc-silicate matrix (Figs. 8–9). This matrix is calcite rich but also contains totally serpentinized grains of metamorphic olivine (or perhaps monticellite) and Al-rich diopside of the type found in altered carbonates and skarn surrounding other ultramafic intrusions (Harris and Chaumba, 2001; Wenzel et al., 2002). Samples LS112 to LS114, taken from within the intrusion at distances of 1 to 5 m from the contact, have normal relict magmatic textures and mineral assemblages (abundant, partially serpentinized olivine, pyroxene, and plagioclase) but are distinguished from other samples by their calcite-rich matrix.

Samples of ultramafic rock collected from more than 5 m from the contact (Fig. 6), and samples of marble only a few tens of centimeters from the contact, reveal little evidence of interaction. In the marbles, mineral assemblages range from

nearly pure calcite, through (serpentinized) olivine-calcite $\pm$ clinozoisite $\pm$ tremolite, to one sample (LS120) that contains about 10 vol percent of white mica. No dolomite was identified in SEM studies. In one sample hydrogrossular has developed as small grains in late fractures that cut across all other minerals, including serpentine after olivine. The mica-rich sample comes from a distinct layer that we interpret as a relict sedimentary bed; the mica-rich assemblage is thought to be the metamorphosed product of impure limestone rich in detrital clay minerals. The olivine in the marbles is interpreted as the product of high-degree, upper amphibolite-facies metamorphism of impure (chert or quartz bearing) originally dolomitic limestones. Subsequent greenschist-facies metamorphism transformed the olivine to serpentine and introduced clinozoisite and tremolite into the carbonate matrix.

In places along the contact, the textures and contact relationships are interpreted to indicate that phenocryst-charged magma was injected into, and mingled with, the carbonate wall rocks. At one location ($38^{\circ}28'44''$ N, $102^{\circ}10'15''$ E), the marble contains patches or isolated grains of now serpentinized olivine with the size and form of phenocrysts in the intrusion (Fig. 8), a texture that suggests that the marble may have been molten.

Southern hanging-wall contact

South of the intrusion the hanging wall mainly consists of orthogneisses, paragneisses, and amphibolites intruded by many amphibolite dikes and granite veins. The gneisses contain many relict layers and lenses of amphibolite. In Mining District II (Figs. 1, 3), the contact between the intrusion and the gneisses is lined with discontinuous slivers of marble, commonly a few meters or tens of meters wide and locally up to 200 m, and rarely of quartzite or quartz veins, up to 30 m wide. The gneisses are very highly deformed and strongly metamorphosed. Many are migmatitic and contain 10- to 30-cm-wide isoclinally folded granite veins. South of the gneisses, biotite amphibolites up to several tens of meters thick are heavily intruded by granitic veins that in places reach 50 vol percent of the rock; these range from deformed granitic melt veins to late homogeneous pink-red granite sheets.

We were not able to demonstrate that the southern contact was intrusive. Where exposed, it is separated from the intrusion by shear zones, the thickness of which varies from a few centimeters to a few meters. Individual displacements are up to about a meter, but it is not possible to relate shear zone thickness with amount of movement, and thus it is not possible to demonstrate how far the gneisses have been transported over the intrusion.

The southern contact is gently curved and its orientation follows that of the foliation of the gneisses. Near the contact, the ultramafic rocks are cut by chloritic shear zones up to 1 m wide. Again, the intrusion appears to have been deformed along with the country rocks. The implication is that the form of the intrusion, in plan and by implication in section, was influenced by postmagmatic deformation. This view is confirmed by the fact that sections based on underground mine data and boreholes show that the main “intrusion” is in fact a large lens, a form that can best be interpreted as tectonic (Fig. 1).

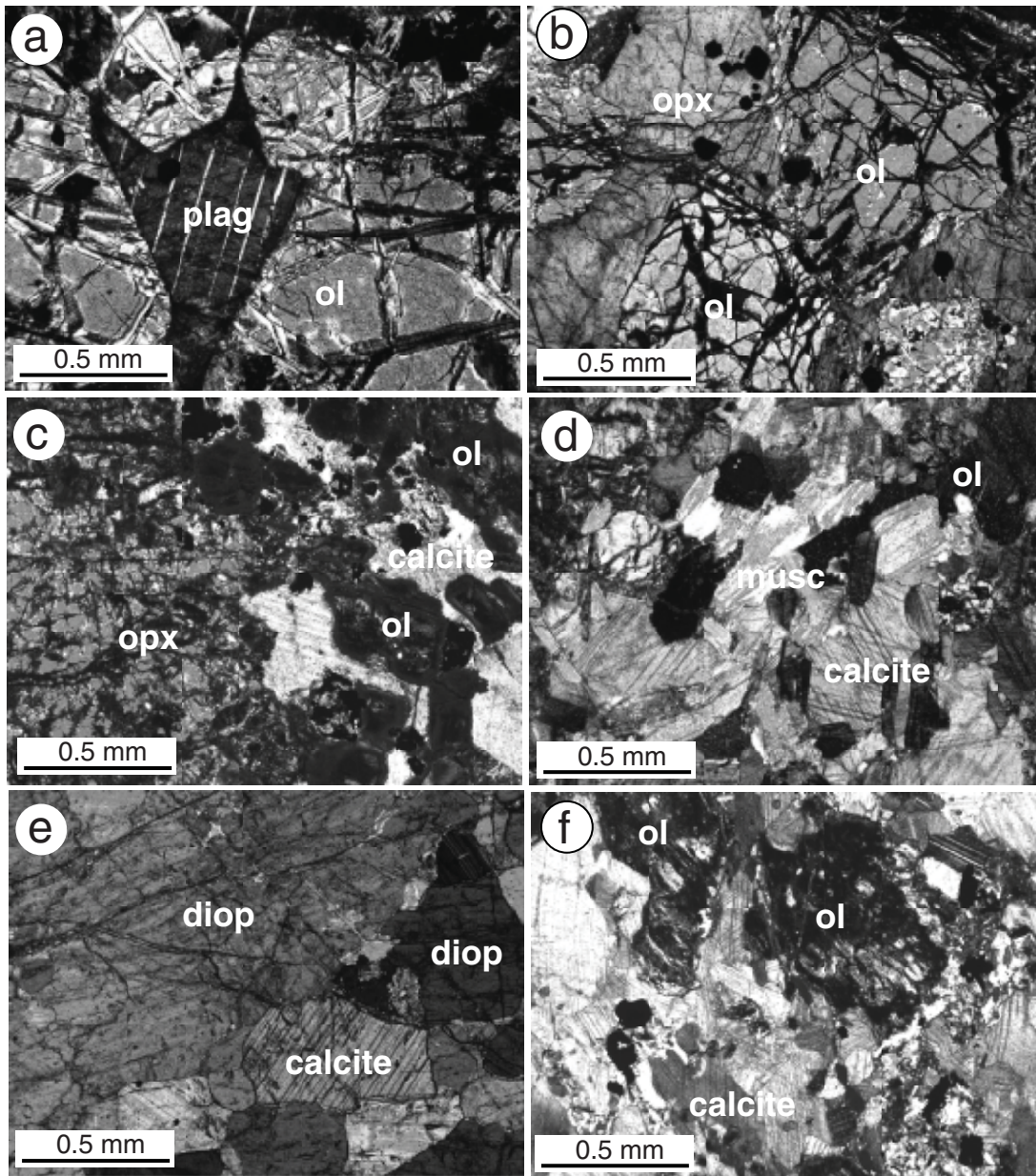


FIG. 9. Photomicrographs, taken with crossed polars, of thin sections from the intrusion and wall rocks. (a). Olivine cumulate with interstitial plagioclase from the southern margin of the intrusion (sample LS137). (b). Olivine cumulate with interstitial orthopyroxene sampled about 30 m from the southwestern contact (sample LS135). Note the magmatic textures and minerals, which are well preserved in this sample. (c). Orthopyroxene crystals, probably magmatic, in a matrix of metamorphic olivine (serpentinized) and calcite (sample LS114). The sample was collected 50 cm from the contact with marble in the map area shown in Figure 6. (d). Muscovite-olivine marble developed from a marly protolith, also from the map area shown in Figure 6 (sample LS120). (e). Diopside skarn in an (originally) marble xenolith at the southern margin of the intrusion (sample LS101). (f). Serpentinized olivine marble adjacent to the contact with the intrusion (sample LS115). Abbreviations: diop = diopside, musc = muscovite, ol = olivine, opx = orthopyroxene, plag = plagioclase.

Deformation and Conditions of Emplacement of the Intrusion

Structural relationships of the Jinchuan intrusion are somewhat similar, although at a larger scale, to those of the boudins of ultramafic rock in the Tamazigou Valley. Although the latter intrusions were emplaced in the Tamazigou Formation, which, according to local geologists overlies the Baijiazui

Formation that hosts the Jinchuan intrusion, a parallel can be drawn particularly to the postintrusion deformation history. On the northeast side of the Jinchuan intrusion, 5 m from the contact (38°28'44 N, 102°10'08 E), the marbles contain broken lenses of amphibolite associated with 1-m-wide sigmoidal shear zones that indicate **thrusting to the north**. At the marble-intrusion contact in Mining District II (38°28'24 N, 102°20'2 E), a 6-m-wide deformation zone of biotite-rich

fault gouge contains thrust-sense kinematic indicators. On the southwest side, the gneiss-intrusion contact is highly sheared. Schists and gneisses dip moderately to the south and are structurally on top of the intrusion. Along the contact, 2- to 5-m-wide amphibolitic schists contain 15-cm-wide, south-dipping shear zones with sigmoidal foliation that indicates **thrusting to the north**, and 1-m-wide shear zones in biotite schist contain relicts and porphyroclasts of brecciated and mylonitized gneiss with thrust-sense kinematic indicators. The thrust relationships are well displayed at the eastern termination of the intrusion where it thins down to several ultramafic lenses a few meters wide within a 5-m-wide shear zone lined with chlorite schists between marbles and gneisses. Sigmoidal drag on shear zones clearly indicates thrusting to the north. **We conclude that the intrusion was emplaced along a contact where gneisses had been thrust over marbles, and that thrust deformation continued to break up the intrusion into lenses on a variety of scales.**

This example of a gneiss-marble contact intruded by the Jinchuan intrusion is not the only one in the region. The 1:50,000 scale geologic map (J48E 010001) shows that south of the Jinchang (the town next to the Jinchuan mine), seven belts of marble alternate with six belts of gneiss, each of which is several hundred meters wide. The Jinchuan intrusion is situated along the northernmost gneiss-marble contact that we have interpreted as a thrust.

Further information comes from two detailed sections across the contact in Mining District II. Figure 10 is a section across the main marble-peridotite contact, located near the road that runs south from Jinchang near section 13 (38°28'52 N, 102°10'02 E; Fig. 3). Two generations of dolerite dikes intruded the marble; the first is sheared by thrusts, and the second is unsheared and was emplaced along the thrusts. There is only one generation of unsheared dikes in the peridotite. We infer that the peridotite intruded along a first generation dike, possibly during intermittent thrusting that sheared it. Thus the intrusion was emplaced between the two generations of dikes. The intrusion itself is cut by a thrust along which there are marble xenoliths some of which are associated with a zone of hybrid olivine-carbonate rock.

Figure 11 is another vertical section across the gneiss-marble contact near 38°28'17 N, 102°10'39 E, where the intrusion has thinned to only a few meters width. The peridotite has been intensely sheared to chlorite schist along a 1-m-wide, north vergent thrust. There is a progression from sheared peridotite, to chlorite schist with sheared peridotite, to chlorite schist with no relicts of peridotite. This is a duplex that also involves slices of sheared marble and granite-gneiss; these relationships provide clear evidence that the intrusion, at least at its margins and extremities, underwent strong postintrusive deformation. Where the intrusion is widest and most rheologically resistant, it is cut by thrusts near its margins; the interior, however, remained relatively undeformed as shown by the preservation of relict magmatic textures and minerals. Where thin and weakest, as at the southeast margin of Mining District II, it is highly deformed throughout and metamorphosed to a greenschist-grade chlorite schist.

All these rocks are located on the southern margin of the North China craton. We infer that the intrusion was emplaced into a shallow-dipping, imbricated shelf (marble and minor quartzite)-gneiss (basement) thrust pile characteristic of the margin of the craton.

Petrography of the Intrusion

The petrographic features and modal mineralogy of selected samples are summarized in Table 1. Our study focused on mine section 18, which crosses the widest part of the intrusion in Mining District II, immediately south of the town of Jinchang. Our emphasis was on the contact relationships. For additional detailed information on the petrography of the intrusion the reader is referred to the earlier and more extensive studies by Chai and Naldrett (1992a, b), Tang (1993), Barnes and Tang (1999), Tang and Li (1999), C. Li et al. (2004), and de Waal et al. (2004). In brief, the intrusion is mainly composed of monotonous olivine-rich cumulates, and pyroxene- or plagioclase-rich facies are developed only at the margins. All rocks are charged with large, near-equant, subrounded grains of olivine (Fig. 4), the forsterite contents of which range from 79 to 86. C. Li et al. (2004) attributed the more Fe rich compositions to interaction between olivine and

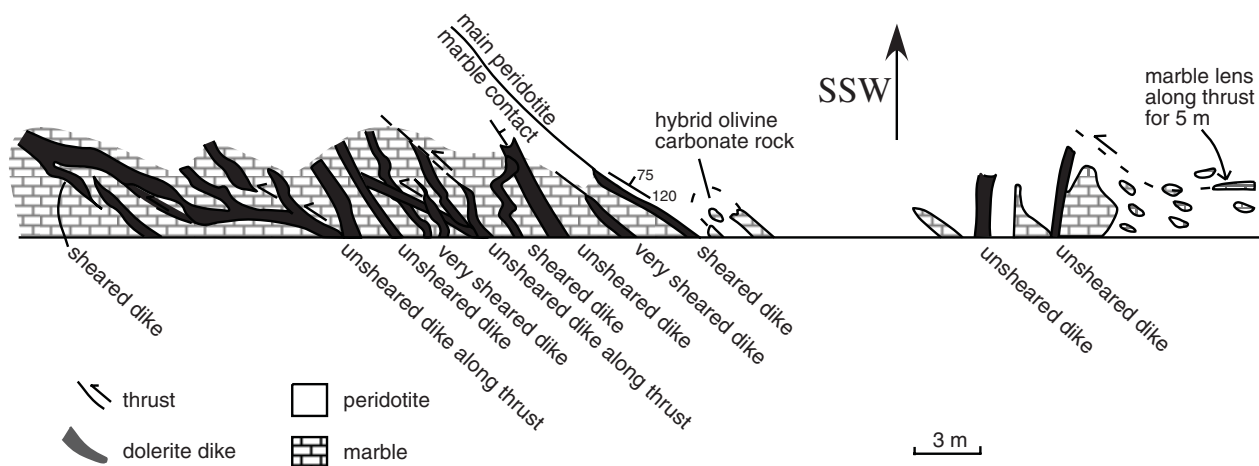


FIG. 10. Vertical section across the main marble-peridotite contact, located at 38°28'52 N, 102°10'02 E near the road that runs south from the town of Jinchang near mine section 13 (Fig. 3).

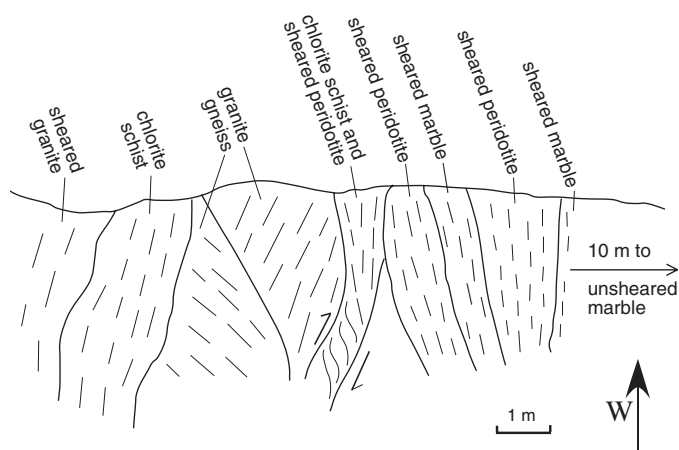


FIG. 11. Section across the southeastern end of the Jinchuan intrusion in mining Mining District II, at 38°28'17 N, 102°10'39 E. The section lies below and to the right of the part of the intrusion shown in Figure 3. The ornamentation indicates the direction of foliation in each of the fault-bounded blocks.

trapped silicate or sulfide liquid. Orthopyroxene, where preserved, is bronzite with a similar Mg/Fe ratio to that of the olivine; the clinopyroxene is augite, and the plagioclase has a high anorthite content.

Secondary minerals are chlorite, tremolite, serpentine, calcite, and oxide belonging to the greenschist-facies metamorphic assemblage (Figs. 4, 8). The minerals in the intrusion and in the wall rocks provide evidence of two stages of

metamorphism. The first, at upper amphibolite facies, affected only the wall rocks and, since the granite dike dated at 2 Ga by Tang and Li (1995) cuts the metamorphic fabric, this metamorphism predated the ~827 Ma intrusion. The second, at greenschist facies, affected the whole Jinchuan belt; this metamorphism transformed the shales of the cover sequence into chlorite schists, retrograded assemblages in the high-grade gneisses and marbles, and partially or completely replaced magmatic minerals throughout the Jinchuan intrusion. In this respect we agree with C. Li et al. (2004) and de Waal et al. (2004), and question the assertion by X. Li et al. (2004, 2005) that the intrusion is largely unmetamorphosed.

The marble xenoliths along the northern contact now consist of a dense mat of millimeter-sized anhedral grains of diopside with minor, much smaller, grains of spinel. Cleavage in the diopside is not deformed, suggesting static crystallization. In places a thin rim of actinolite or tremolite surrounds the xenoliths. Magmatic olivines in peridotite immediately adjacent to xenoliths have compositions similar to those in regions without xenoliths but the matrix between these olivine grains is altered to amphibole and chlorite.

Geochemical Investigations

Compositions of rocks from the intrusion (recalculated to 100% on a volatile-free basis) are plotted in Figure 12, along with measured compositions of magmatic minerals. Using an olivine composition of Fo₈₄, Chai and Naldrett (1992a) calculated that the parental silicate liquid of the intrusion contained about 12 percent MgO, a composition also accepted by

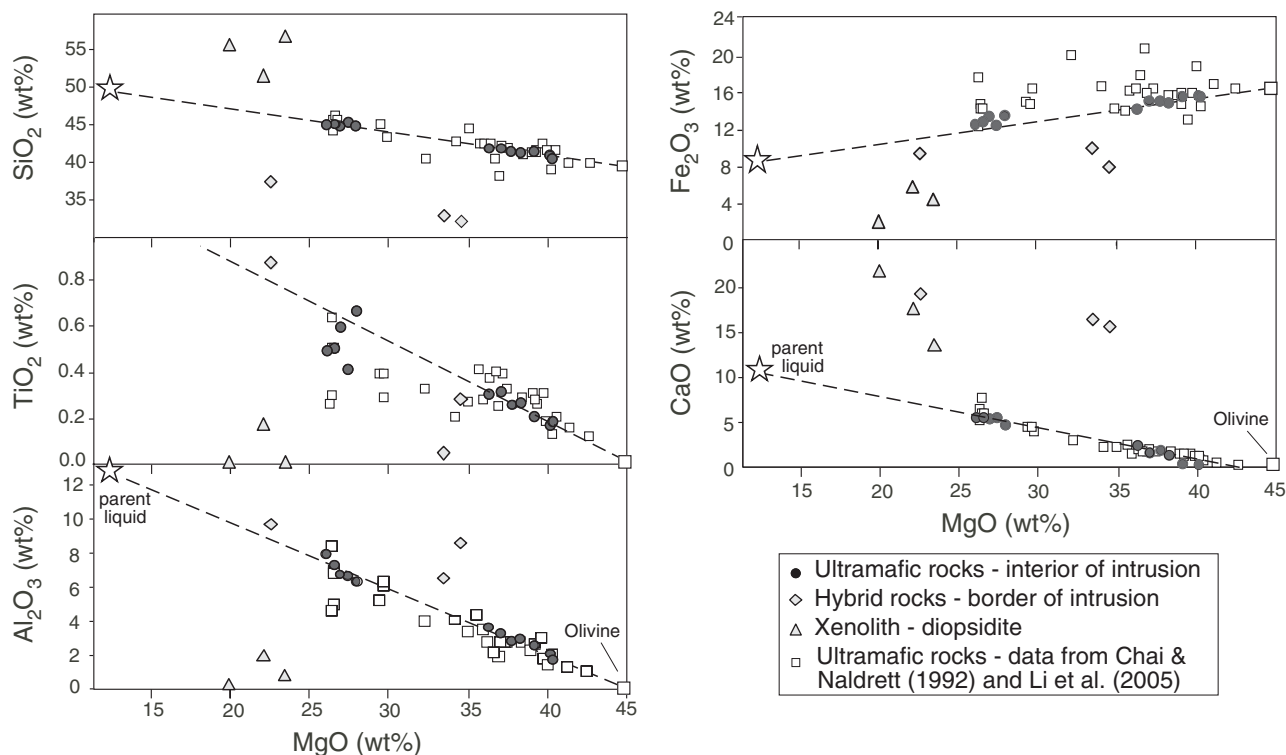


FIG. 12. Variations in major element compositions of rocks from the intrusion (data normalized to 100% on a volatile-free basis). The star shows the composition of the parental liquid, as estimated by Chai and Naldrett (1992a); the open square shows the composition of the most abundant olivine. Unaltered samples lie between these compositions, showing that the rocks are olivine cumulates or mushes.

C. Li et al. (2004). This composition and that of Fo₈₄ are plotted in the diagrams. The compositions of most of the intrusive rocks plot along well-defined linear trends between the two compositions (see, for example, MgO vs. SiO₂ and MgO vs. Al₂O₃ in Fig. 12) confirming the conclusion that only olivine, and not pyroxene and plagioclase, was involved in the magmatic crystallization of the intrusion.

Three samples from the margin of the intrusion plot off the main trend. These samples (LS112, LS113, and LS114) have abnormally low SiO₂ contents and abnormally high CaO and loss on ignition, and in Figure 13 their compositions (not normalized and including LOI in this diagram) plot midway between those of the majority of rocks from the intrusion and the composition of the marbles. This finding, together with

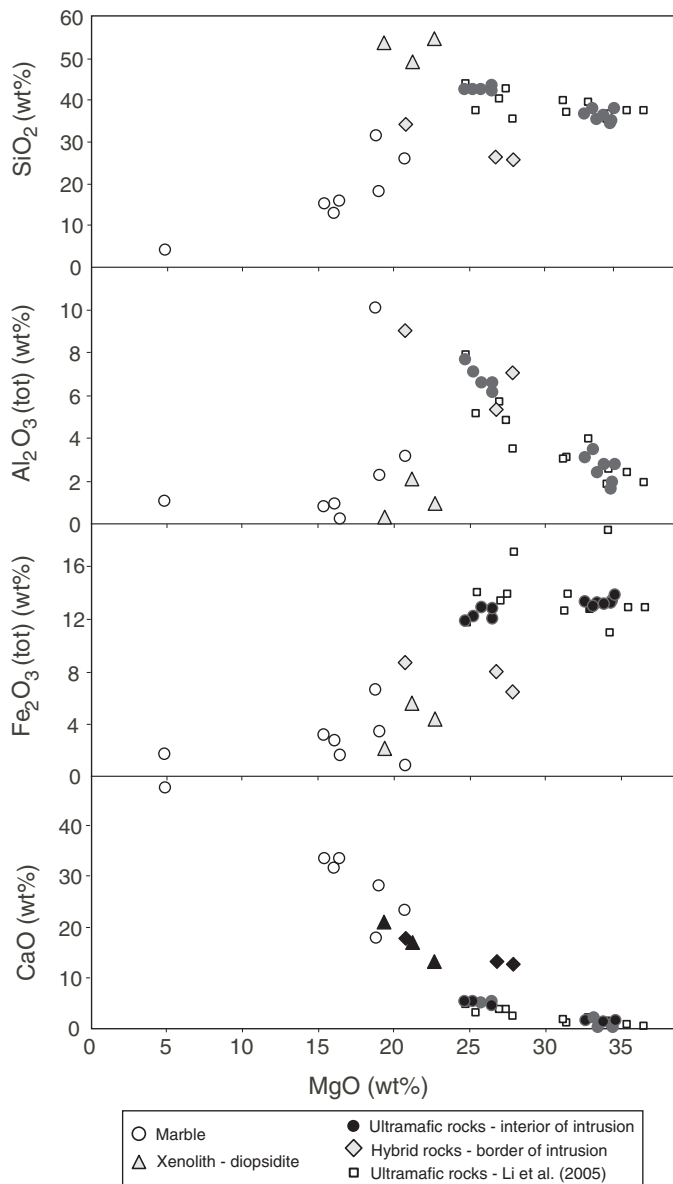


FIG. 13. Variations in major element compositions of rocks from the intrusion and from adjacent marbles (not normalized). The hybrid rocks plot between the compositions of unaltered samples and marbles, except in the MgO vs. Al₂O₃ diagram.

the high proportion of calcite in the matrix of these rocks, confirms that the rocks are hybrids, the products of interaction between magma and marble. Additional evidence comes from the oxygen isotope compositions. Figure 14 shows the correlation between CaO/SiO₂ and whole-rock $\delta^{18}\text{O}$, two sensitive measures of contamination of magmatic rocks by marble. The three hybrid rocks have relatively high CaO/SiO₂ ratios and $\delta^{18}\text{O}$ values, and their compositions plot between 5 and 6 per mil, the value of mantle-derived magma (e.g., Eiler, 2001) shared by the majority of rocks from the intrusion and the higher values of the marbles. Although the positions of the samples in these diagrams conform well to a simple mixing relationship, the position of the hybrid rocks in the MgO versus Al₂O₃ diagram (Fig. 12) shows that the situation is more complicated. The Al₂O₃ contents of the rocks in question are collinear with those of other, apparently uncontaminated, intrusive rocks and they do not contain the lower Al₂O₃ contents that would be expected if the peridotite had been diluted with Al₂O₃-poor carbonates. It appears that the change in CaO and SiO₂ was not due simply to the introduction of calcite into the intrusive rocks, which should have had similar effects on both SiO₂ and Al₂O₃, but involved more subtle interaction between carbonate-bearing fluids and the magmatic rocks.

Figure 15 shows that when $\delta^{18}\text{O}$ of the calcite is plotted against the oxide contents of the whole rocks, there are good negative correlations with SiO₂ and MgO and a positive correlation with CaO. This excludes simple mixing of carbonate and silicate material and is also inconsistent with low-temperature replacement by carbonates. A correlation between calcite $\delta^{18}\text{O}$ and whole-rock oxide contents could only occur if there was exchange between the silicates and calcite at relatively high temperature, either during metamorphism and/or during fluid-rock interaction. Hence where the silica content is high, exchange between the calcite and silicates shifts the calcite $\delta^{18}\text{O}$ toward that of the silicates. Regardless of the exact mechanism, the data suggest a high-temperature process not mechanical mixing or low-temperature replacement by calcite.

There is no correlation between $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$, and $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ values show no systematic variation with the proportion of calcite in the sample. This lack of correlation suggests

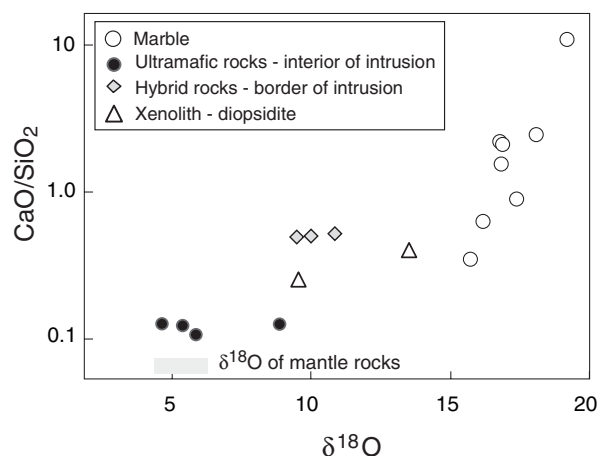


FIG. 14. Correlation between whole-rock oxygen isotope compositions and the CaO/SiO₂ ratio of rocks at the border of the intrusion, showing clear evidence of the assimilation of marble.

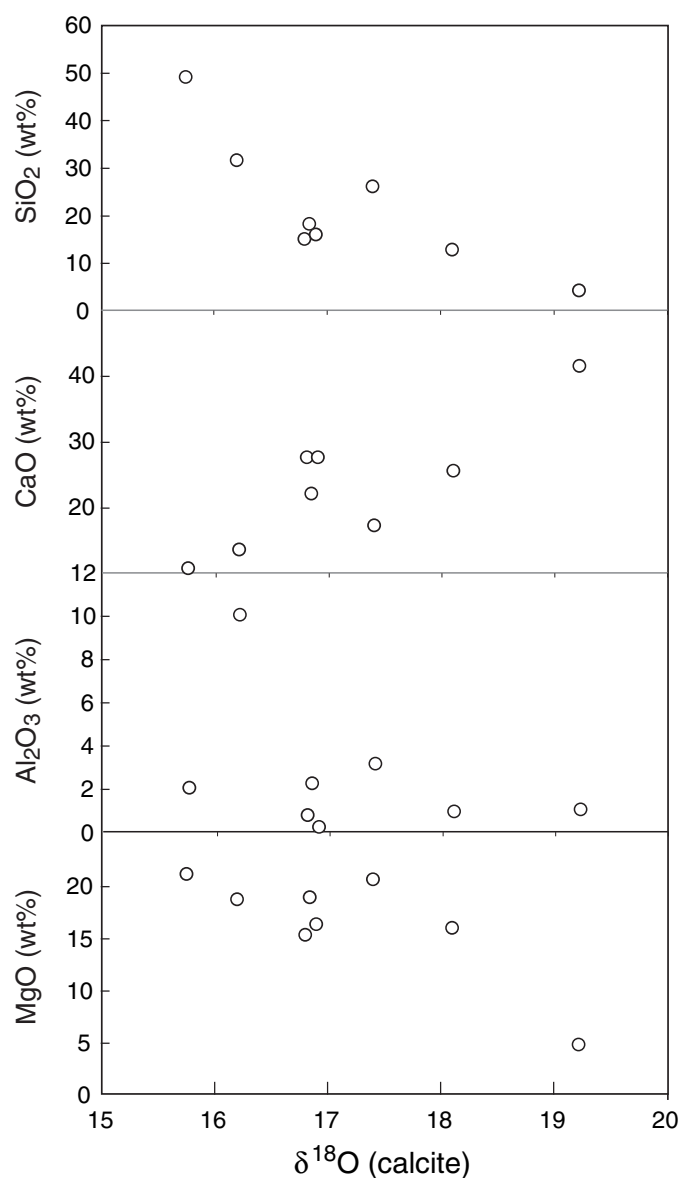


FIG. 15. Correlations between the $\delta^{18}\text{O}$ of calcite and major oxide contents of whole-rock samples, showing that the extent of carbonate assimilation may be related to the rock composition.

that degassing of CO_2 during decarbonation had little influence on the carbonate stable isotopes. On the other hand, one of the xenolith samples has much more negative $\delta^{13}\text{C}$ and low calcite content, and this is consistent with decarbonation. Thus, the interaction between the ultramafic rock and carbonate was a high-temperature process.

Figure 16 shows the correlation between selected trace elements and MgO. In the case of immobile elements such as Th, Nb, Zr, and the rare earth elements (REE), samples from the intrusion again lie on olivine control lines. Figure 17 is a plot of the mantle-normalized trace element patterns of intrusive rocks and marbles. The trace element contents of the marbles are low, in most cases similar to those of the intrusive rocks. Although the patterns are irregular, most elements are not present in anomalously low or high concentrations. A

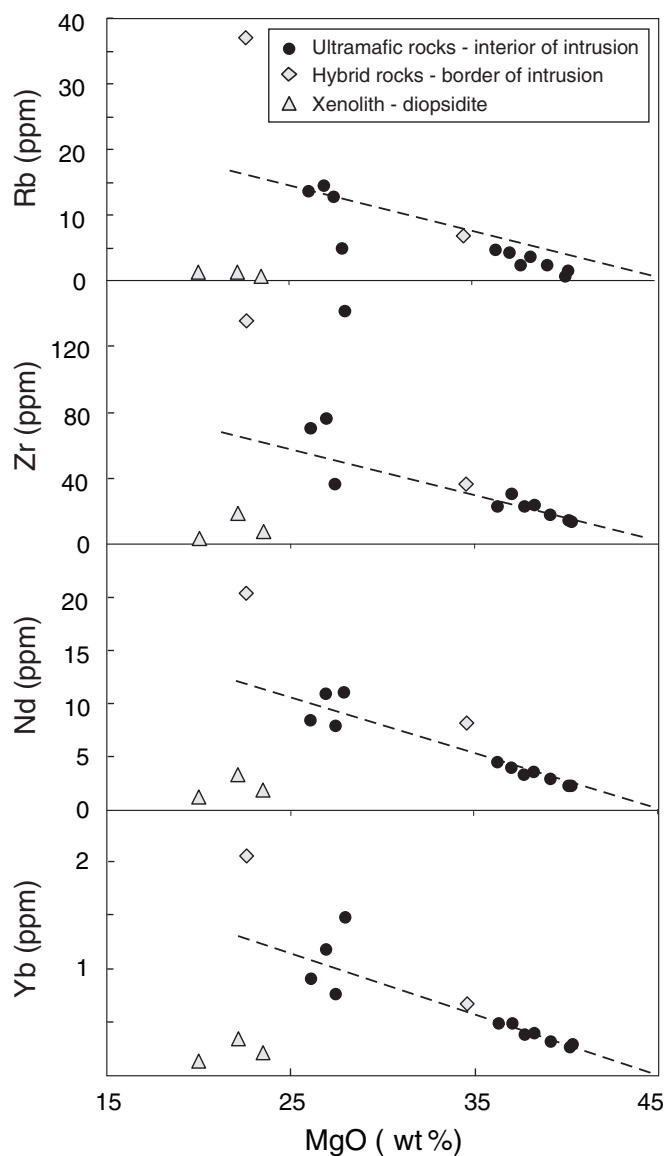


FIG. 16. Variations of selected trace elements, plotted against MgO. The data plot on or near the olivine control lines (dashed).

possible exception is U, which is relatively enriched in the marbles and skarn xenoliths; it is also enriched in some of the samples of ultramafic rock closest to the contact, particularly those whose major element and oxygen isotope compositions record interaction with carbonate-bearing fluids.

The mica-rich marble, sample LS120, contains significantly higher concentrations of Al_2O_3 , SiO_2 (Fig. 12), and K_2O , Cs, Rb, and Ba (Table 1) than the other marble samples, consistent with the original presence of detrital clay minerals.

Concentrations of incompatible elements are slightly higher in samples of ultramafic rocks from the border of the intrusion than in samples from the interior (Figs. 12, 17). This difference results from a high proportion of trapped liquid, now expressed as intercumulus pyroxene or plagioclase, in samples from the border zones.

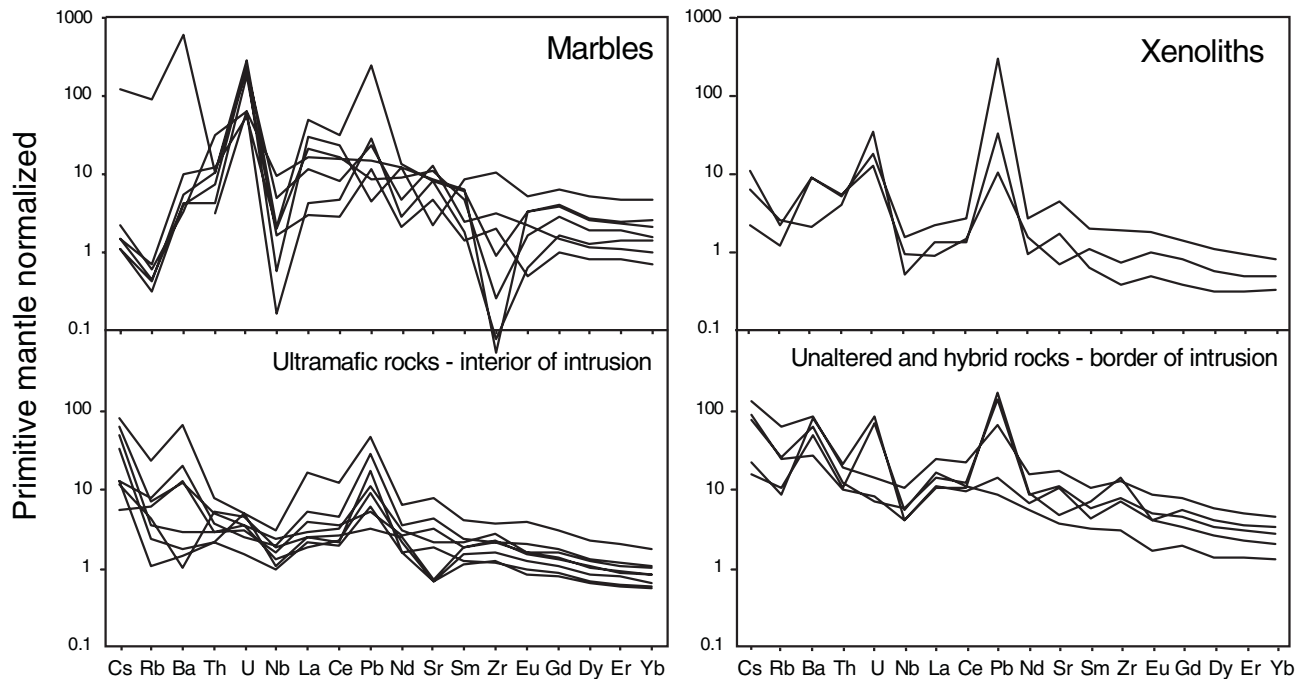


FIG. 17. Trace element concentrations, normalized to the primitive mantle of Hofmann (1988), of ultramafic rocks from the intrusion, some hybrid rocks and xenoliths and marble from the northern margin.

The mantle-normalized trace element patterns in the intrusive rocks themselves are distinctive (Fig. 17). Virtually all samples show relative enrichment of the more incompatible elements (e.g., high La/Yb), large positive Pb anomalies, and large negative Nb anomalies. We interpret these characteristics, which are typical of granitic or sedimentary rocks from the continental crust (e.g., Taylor and McLennan, 1985), to indicate that the rocks of the intrusion are highly contaminated with crustal material. All the samples have near-parallel mantle-normalized trace element patterns and the overall concentrations of these elements correlate with MgO contents (Fig. 16). This correlation is consistent with a model in which chemical variation within the intrusion is controlled essentially by the amount of cumulus olivine of relatively constant composition; i.e., FO_{84} , as recorded in microprobe analyses (Table 1; Chai and Naldrett, 1992a; C. Li et al., 2004) or as inferred using the major element data. In addition, the identical slope of the patterns of all samples indicates that they were all contaminated to about the same extent. We conclude that the contamination took place before the magmas entered the magma chamber or conduit that is now preserved as the Jinchuan intrusion.

Taking the argument further, the normalized abundances of Th and U in the ultramafic rocks plot below those of light rare earth elements (LREE) like La and Ce (Fig. 17). Although U is mobile during alteration, the relatively constant ratio of this element to less mobile elements such as Th and the REE (except in the marble-contaminated samples) suggests that its abundance has been little influenced by metamorphism. A low abundance of Th and U compared with the LREE is characteristic of rocks from the lower continental crust (e.g., Rudnick and Fountain, 1995). Therefore, we conclude that

the magma that fed into the Jinchuan chamber had become highly contaminated in the lower part of the crust.

Origin of Jinchuan Magmas

If the primary magma that fed the Jinchuan intrusion had become highly contaminated with continental crust, as we infer from the trace element characteristics of samples from the intrusion, it is very difficult to establish its original chemical characteristics. Given that even after strong crustal contamination the magma that fed into the intrusion retained a high MgO content (12% according to Chai and Naldrett, 1992a), the primary magma must have had a much higher magnesian composition, most probably that of a picrite. We propose that the primary magma that fed the Jinchuan intrusion formed through high-degree partial melting of an unusually hot mantle source, i.e., a plume according to most authorities (e.g., Campbell and Griffiths, 1990). This interpretation is not consistent with the hypothesis of Li et al. (2005) that the magma was derived from the subcontinental lithospheric mantle. These authors based their argument on the relative enrichment of the more incompatible trace elements, but this feature could have resulted either from low-degree melting under conditions that left garnet in the melting residue or from the crustal contamination.

The primary magma could originally have been depleted in incompatible trace elements, like the picrites and komatiites of Gorgona Island (Arndt et al., 1997), or it could have been enriched in these elements like the picrites of the Duarte Formation, another part of the Caribbean oceanic plateau (Lapierre et al., 1999). Given that our geochemical data on the magmatic system are currently limited to those of samples from the small, highly deformed and heavily contaminated

Jinchuan intrusion, we have no way of determining the exact nature of the primary magma. Similarly, the amount of contamination is impossible to calculate accurately given our lack of knowledge of the compositions of both the primary magma and contaminant(s) and of the contamination mechanism (i.e., bulk contamination or assimilation of partial melt of the crust or contamination in closed- or open-system magma chambers). A simple model that assumes bulk assimilation, a parental magma of picritic composition, and a contaminant with the composition of Rudnick and Fountain's (1995) lower continental crust indicates between 10 and 30 percent contamination. Radiogenic isotope analyses currently underway will provide additional constraints.

Despite these uncertainties, two important conclusions stem from the geochemical investigation. The primary magma that fed the Jinchuan intrusion was highly magnesian, and this magma became highly contaminated in the lower continental crust before it entered the intrusion.

Emplacement of the Intrusion

The wall rocks consist of marbles to the north of the intrusion and felsic gneisses to the south. These rocks had undergone partial melting and their mineral assemblages correspond to upper amphibolite facies. The Neoproterozoic sediments of the cover above the unconformity were metamorphosed under greenschist-facies conditions. Granitic and pegmatitic dikes cutting rocks surrounding the intrusion are dated as Paleoproterozoic. These dikes lack textures and mineral assemblages corresponding to amphibolite facies and cut across the metamorphic fabric; therefore, they are interpreted to postdate the high-grade metamorphism. X. Li et al's (2004, 2005) zircon dating indicates that the intrusion is Neoproterozoic and therefore much younger than the granitic and pegmatitic dikes; if so, it must also postdate the high-grade metamorphism. We can conclude on this basis that the Jinchuan magma intruded a sequence of already metamorphosed gneisses and marbles.

The 1:50,000 geological map (J48E 010001) shows that the intrusion is currently only a few kilometers from the unconformity at the base of the cover sequence, and although this distance might have been reduced during the deformation that affected the entire belt, the change may not have been great. The implication is that the intrusion could have been close to the surface when the Upper Proterozoic sediments were deposited; in other words, the level of emplacement may not have been the middle to lower crust but rather in a metamorphic terrane that had been brought close the surface by erosion and uplift.

Emplacement of the intrusion could have taken place before, during, or after deposition of the Neoproterozoic cover sequence but before the greenschist-facies event, because this event overprinted the intrusion itself and deformed its margins. It is difficult to establish whether the emplacement occurred before or after tilting of the unconformity. It is conceivable that the magma was emplaced as a near-vertical unit along lithologic contacts within an already tilted, steeply dipping gneissic sequence. However, we accept the interpretation of de Waal et al. (2004), who used the lithologic variations within the intrusion and the disposition of the orebodies to argue that the unit originally had a near-horizontal orientation,

and propose that intrusion preceded rotation of the unconformity to its present near-vertical orientation. We further conclude that, following contamination in the lower crust, the magma intruded as a sill into a series of amphibolite-facies gneisses and marbles. It solidified and then was deformed together with the high-grade host rocks, subjected to retrograde metamorphism during erosion and uplift, remetamorphosed and deformed once again under greenschist-facies conditions during deposition and burial of the Neoproterozoic cover sequence.

A New Model for the Jinchuan Deposit

Several observations form the basis of a new model for the formation of the Jinchuan Ni-Cu-(PGE) deposit.

1. Inclusions of marble, now decarbonated and converted to diopside-rich skarn, are common near the contact and dispersed through much of the intrusion. In contrast, xenoliths of felsic gneiss appear to be absent.

2. Metasomatism of the marbles seems to be restricted to a narrow aureole along the base of the intrusion. Although the underlying marbles contain abundant olivine and in places, mica, these are interpreted as the products of metamorphism of impure dolomite. Skarn minerals such as Al-rich diopside are restricted to a thin irregular zone within a few tens of centimeters of the contact.

3. The immediate base of the intrusion consists of olivine cumulates with carbonate-rich matrices, and marble immediately below the contact contains large rounded relict olivine grains (now serpentine). The relict olivine grains have the size and form of cumulus grains in the intrusion (Fig. 4) and are very different from the much smaller, euhedral metamorphic olivines. These observations suggest that marble was mechanically assimilated into the intrusion, and alteration of the rocks above and below the contact by processes such as fluid circulation was very limited.

Most magmatic Ni-Cu-(PGE) sulfide deposits form when magma assimilates its wall rocks (Naldrett, 1992, 2004). This process decreases the solubility of sulfide in the silicate liquid, if siliceous rocks are assimilated, or increases the sulfur content of the silicate liquid, if sulfur-rich rocks are added. The rocks surrounding the Jinchuan deposit are very poor in sulfur and, therefore, the latter ore-forming process appears to be ruled out. Sulfide segregation triggered by the assimilation of siliceous granitoid rocks remains a possibility, particularly in view of the trace element evidence that the Jinchuan rocks are heavily contaminated with material from the lower continental crust. It is conceivable that the sulfides that now constitute the Jinchuan orebodies segregated at depth and were subsequently transported to their present level in an olivine crystal-charged magma (e.g., de Waal et al., 2004; Tang and Li, 1999). The problem with such a model lies with the difficulty of transporting upward a magma consisting of olivine crystals and sulfide, a mixture with density far greater than that of the wall rocks. In this case, normal emplacement of magma, which is essentially driven by magma buoyancy, would have to be ruled out, and some type of tectonic squeezing must be invoked.

De Waal et al. (1993) suggested that differentiation of olivine crystals, sulfide liquid, and silicate liquid took place in

a staging chamber at a depth between 4 and 9 km and that magma charged with olivine crystals and sulfide droplets was squeezed out by collapse of the chamber roof during a period of regional extension. Although this might be a plausible mechanism if the magmas were emplaced during a period of interthrusting extension, our independent evidence for a close association between magma emplacement and thrusting suggests that a more likely and viable mechanism was magma injection driven by compression of the crust during ascent of the magma.

In our reconstruction of the setting of the Jinchuan deposit, the host intrusion was emplaced along a thrust contact between a series of marbles, which formed the footwall and gneisses and minor marble, which formed the hanging wall. The northern (lower) contact with marbles is sharp but undulating and appears to be magmatic. In places, the crystal-charged magma invaded the floor rocks, and xenoliths of marble, now transformed to diopside-rich rocks, were incorporated into the lower part of the intrusion. Chemical data, principally major element and oxygen isotope compositions (Figs. 12–16), provide clear evidence that the lower border of the intrusion assimilated marble. Xenoliths within the intrusion, and some rocks along the lower contact, are now partially to completely decarbonated, the original mineralogy of the marbles having been replaced by diopside and other silicates. It appears, therefore, that during its emplacement, the lower part of the intrusion interacted with the floor rocks, and, crucially, this part of the intrusion was invaded by CO₂-rich fluids released by decarbonation of the uppermost marbles and of marble xenoliths. These fluids would have percolated up into the crystal mush.

Haughton et al. (1974) and Buchanan and Nolan (1979) demonstrated that under certain conditions, sulfide solubility correlates inversely with oxygen fugacity. They concluded that an increase in f_{O_2} might also lead to the separation of an immiscible sulfide liquid. Although other studies have given conflicting results (Mavrogenes and O'Neill, 1999), theoretical arguments favor the former interpretation (see the discussion by Naldrett, 2004, p 21–26). In silicate magma under moderately reducing conditions appropriate for mantle-derived magmas, the sulfur solubility is directly linked to the iron content or, more precisely, to the Fe²⁺ content. An increase in oxygen fugacity will change Fe²⁺ to Fe³⁺, reducing the effective Fe²⁺ content of the magma and thus decreasing the sulfur solubility.

A role for CO₂ is supported by investigations of two other mafic-ultramafic bodies that intruded carbonate sequences. The first is the Ioko-Dovyren intrusion in eastern Siberia, which was recently studied by Wenzel et al. (2002) and which provides key information about how mafic magma invades and assimilates calcareous sedimentary rocks. The second is the northern limb of the Bushveld Complex, which, in the Platreef region, intrudes dolomitic limestones. In this region, the interaction between mafic magma and carbonates seems to be associated spatially, and perhaps genetically, with the formation of PGE deposits (e.g., Harris and Chaumba, 2001). Wenzel et al. (2002) studied in detail the petrology and mineralogy of xenoliths within the ultramafic intrusion. They concluded that these xenoliths, which now have skarn mineralogy dominated by Al diopside, were blocks of footwall dolomitic

limestone that had become incorporated into the ultramafic intrusion during its emplacement. Wenzel et al. (2002) showed that the liquidus of Mg-Ca carbonate in the presence of water lies well below the solidus of mafic magma, and they postulated that the carbonate wall rocks partially melted to form a low-density, low-viscosity liquid. This liquid penetrated up into the overlying ultramafic intrusion and, as it did so, it was decarbonated and transformed into the skarn assemblage. Crucially in this context, Wenzel et al. (2002) showed that the interaction of carbonate with the ultramafic rock increases the oxygen fugacity of the silicate liquid.

In the Platreef around the Sandsloot PGE mine, siliceous mafic to ultramafic magma (probably komatiite that was heavily contaminated with granitic crust: Sharpe et al., 1986; Maier et al., 2000) intruded a series of dolomites, shales, and granites (Harris and Chaumba, 2001). “Para-pyroxenite,” a diopside-rich skarn, developed extensively along the contact with dolomite. Oxygen isotope data from magmatic pyroxenes indicate that 10 to 30 percent carbonate was assimilated into the mafic magma (Harris and Chaumba, 2001).

We suggest here that the assimilation of carbonate may have played a key role in the formation of the ore deposits at Platreef and Jinchuan. Further, we propose that an increase in oxygen fugacity resulting from the assimilation was a key factor in this ore-forming process. In the Ioko-Dovyren intrusion (Wenzel et al., 2002) and at the Sandsloot mine (Gain and Mostert, 1982), disseminated Ni and Cu sulfides surround skarn xenoliths, showing that the mineralization process works on a small scale. In the Platreef, the best PGE grades on the Platreef and the only operating mine are located where the footwall is carbonate. The Ioko-Dovyren intrusion did not form an ore deposit (the parental magma may have been too far from sulfide undersaturation) or its ore deposits have yet to be found. In the Platreef, the assimilation of carbonate appears to have induced sulfide saturation in a Ni- and S-poor, but PGE-rich magma. A small proportion of sulfide droplets captured PGE from the ultramafic magma and accumulated close to the contact to form the ore deposits.

In Jinchuan, an olivine-phenocryst-charged magma intruded along a contact between marbles and paragneisses (Fig. 18). This magma had earlier assimilated a high proportion of granitic material in the lower crust, a process that is crucial to our model. Mavrogenese and O'Neill (1999) have shown that the solubility of sulfur in mafic-ultramafic magma depends strongly on pressure. An ultramafic magma formed through melting deep in the mantle can dissolve only a limited amount of sulfur, and when this magma reaches crust levels, the pressure drop causes it to become strongly undersaturated in sulfur (Leshner and Keays, 2002). If an ore deposit is to form directly from such magma, either considerable sulfur must be added from an external source or the magma must evolve in other ways. In our model, the hot, primitive magma assimilates lower crustal wall rocks and partially crystallizes, processes that both decrease the sulfur solubility and increase the sulfur concentration of the melt (as crystallization proceeds, the sulfur becomes concentrated in an ever decreasing volume of residual magma). When the magma reached its present level in the mid to upper crust and came into contact with the marbles, we presume it was close to sulfur saturation.

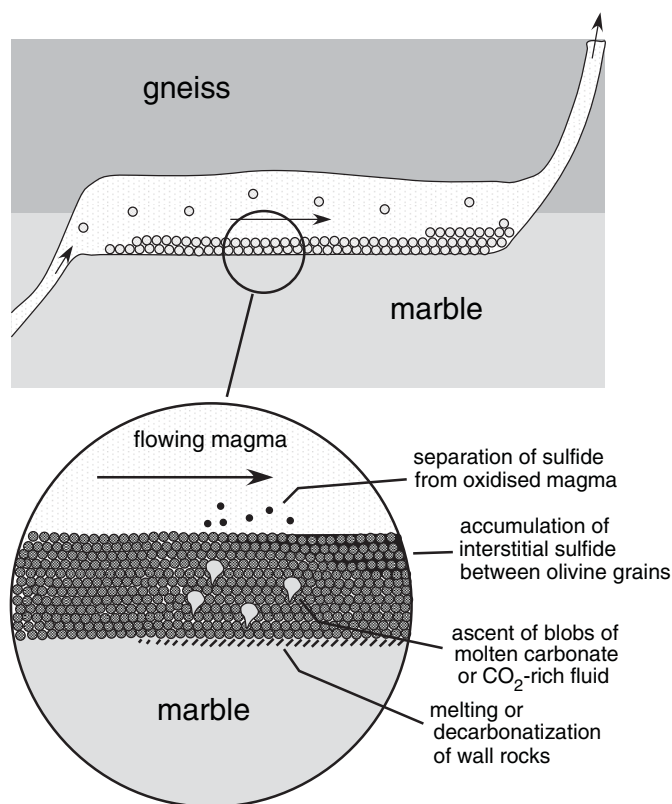


FIG. 18. Sketch illustrating the processes that may have formed the Jinchuan sulfide deposits. See text for explanations.

Heat from the magma may have partially melted the carbonate wall rocks or xenoliths, causing blobs of low-density, low-viscosity, calcite-rich liquid to rise up through the olivine cumulate in the lower part of the intrusion. Alternatively, CO_2 -rich fluids released by decarbonatization of the floor rocks and/or xenoliths could have percolated up through the cumulate. On reaching the cumulate-liquid interface, the fluids reacted with the magmatic liquid, increasing its oxygen fugacity and causing sulfide liquid to segregate. If the magma were moving through the intrusion, the carbonate would oxidize and cause sulfide to segregate from a constantly flowing stream of magma; thus a large volume of sulfide could accumulate in the site of this interaction. The sulfide liquid, being far denser than the silicate liquid, would have percolated down through the crystal mush that formed the upper part of the cumulate pile to just above the level where the rock was completely solid, and it would have accumulated there to form the net-textured ores that characterize the Jinchuan deposits. This mechanism explains the position and the nature of the mineralization of the deposits, as net-textured sulfide in the lower part of the intrusion and not at its basal contact.

Conclusions and Implications

A now vertical disconformity between gneisses of the Longshoushan belt and Neoproterozoic metasedimentary rocks some 20 km west of Jinchuan is parallel to the orientation of the Jinchuan intrusion. Rotating the unconformity to

its original horizontal position brings the intrusion to a near-horizontal orientation, supporting de Waal et al.'s (2002) proposal that the intrusion was a sill, not a subvertical trumpet-shaped body. The intrusion probably was emplaced as a semiconformable body along a thrust between marble and overlying gneisses. After intrusion, the body was metamorphosed, at least to lower greenschist facies, and tectonically extended into a series of large lenses. Despite this complicated history, contact relationships are, in places, well preserved. Although the southern (originally upper) contact with felsic gneisses is sheared, the northern (lower) contact appears to be intrusive. **At a broad scale the intrusion is conformable with the foliation of the surrounding rocks, but locally it is discordant, with strike oblique to relict bedding in the marbles.**

At the level of the intrusion there is no evidence of assimilation of felsic gneiss from the hanging wall but abundant geological and geochemical evidence of interaction with marble from the footwall. The major and trace element data reveal considerable modification of the composition of the ultramafic rocks adjacent to the lower contact. An increase in Ca and a decrease in Si, for example, are attributed to assimilation of carbonates. The trace element patterns of rocks from the intrusion exhibit strong enrichment of incompatible elements, large negative Nb-Ta anomalies, and low U-Th contents, features attributed to the assimilation of granitoid lower crustal rocks. These patterns are present in all samples from the intrusion, indicating that the assimilation took place before the magma was emplaced at its present level in the crust. Because the exposed country rocks are poor in sulfides, we propose that ore formation was linked to the assimilation of carbonate into magma that was close to sulfide saturation as a result of earlier assimilation of granitic crust. Decarbonation of marble liberated CO_2 which penetrated into the intrusion, causing an increase in oxygen fugacity that led to the segregation of ore sulfides.

If this model is correct, it has significant implications for the exploration of deposits similar to Jinchuan. For example, similar deposits might be found in the Emeishan volcanic province in southern China where mafic magmas have intruded calcareous sediments (Zhou et al., 2002). In the Panzhihua intrusion in Sichuan province, there is abundant evidence of interaction between magma and marble (Zhou et al., 2005), and this interaction appears, at least spatially, to be associated with unusual Fe-Ti-V oxide ores. During exploration for such deposits, the geochemical signatures of the assimilation of granitoids or terrigenous sediments are not particularly relevant because they only indicate that the magma was capable of assimilating its wall rocks. More appropriate are indices specific to the assimilation of carbonate, particularly elevated CaO contents and, in unaltered sequences, high $\delta^{18}\text{O}$ values, especially of separated minerals such as pyroxene, which are relatively resistant to secondary exchange with fluids. The use of mineral compositions to estimate directly the oxygen fugacity (e.g., oxide compositions, V in olivine) might also provide valuable information.

This improved understanding of how the interaction of magma with country rocks is linked to ore formation should be included in criteria designed to aid exploration for magmatic ore deposits.

Acknowledgments

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